

Research Report on

Validation methods for scientific ML simulators via physics-based metamorphic testing and relation violation statistics

Full search query: I want to find literature on validation methods for broad scientific machine learning systems—especially PINNs, neural operators, surrogate PDE models, and learned scientific simulators—that connect metamorphic testing, property-based/relation-based oracles, and physics-based V&V, with emphasis on using physically derived out-of-distribution transformations as follow-up tests and using executable relation-level violation rates to characterize the reliable operating boundary of the system under test, while accounting for the domain validity of each relation under governing equations, boundary conditions, discretization, tolerances, and implementation assumptions

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Summary

The search uncovered only a few directly aligned works on physics-based metamorphic / relation-oriented validation for scientific ML—most notably the fluid-surrogate testing frameworks of Qi and Yu [1,2]—while the rest provide complementary but incomplete ingredients (residual-based certification [3,10,11], generic MT for scientific software [4,5,6,9], and residual-threshold operating logic in hybrid solvers [7,8]) that together highlight a clear but still-open research gap around relation-violation-based operating envelopes with explicit domain-of-validity handling.

Overall Assessment of the Landscape

Key point: There is **no existing, fully integrated framework** that does all of the following for PINNs, neural operators, surrogate PDE models, or learned simulators:

- Defines **physics-grounded metamorphic / relation-based oracles** (PDE residuals, similarity laws, symmetries, constitutive relations, etc.) *post-training*,
- Uses **physically derived OOD transformations** of test cases as systematic follow-up tests,
- Quantifies **relation-level violation rates** across input/parameter space and converts them into an explicit **reliable operating boundary**, and
- Explicitly encodes **domain-of-validity for each relation** (governing equations, BCs, regimes, discretization/tolerance assumptions) to separate “out-of-model-domain” from “model failure.”

Instead, the literature offers **four partially overlapping strands**:

1. **Physics-based metamorphic testing for scientific ML flow surrogates** – closest to your target ([1,2]).
1. **Assumption-/physics-based metamorphic testing and MR identification for simulation/CPS software** – methodologically important but not ML-PDE-focused ([4,5,6,9]).
1. **Physics-informed UQ and certification for neural PDE surrogates and PINNs** – strong on residual-based nonconformity and guarantees, not on metamorphic/OOD relations or operating envelopes ([3,10,11]).
1. **Hybrid ML–solver schemes using residual thresholds as dynamic trust criteria** – operationalize residual-based “trust regions,” but without metamorphic tests or explicit regime maps ([7,8]).

What follows focuses on how each strand contributes to, and falls short of, your desired framework.

Closest Existing Realizations of Physics-Based Metamorphic Testing for Scientific ML

Fluid-dynamics surrogates with dimensionless-number-based metamorphic relations – Qi & Yu group [1,2]

Systems under test: “Intelligent algorithms” for **physical field prediction**—practically, ML surrogates for velocity fields around a cylinder (CFD-type problems) [1,2]. Architectures are abstracted away; the methodology applies to generic learned field predictors.

Core ideas that align strongly with your goal:

- **Metamorphic relations grounded in fluid mechanics:**
 - Relations are constructed using **Reynolds number conservation** and **Strouhal number correlations** for cylinder wake shedding [2], and more generally **multi-parameter physical correlations and empirical statistical patterns of key flow parameters** [1].
 - These MRs encode **dimensionless similarity and known scaling relations**, not just ad hoc input-output heuristics.
- **Physically derived OOD / follow-up scenarios:**

- Test design is **scenario-based** with four categories: **benchmark, extreme, perturbation, cross-domain** [1].
- **Disturbance** and **cross-domain** scenarios are explicitly built by **altering flow conditions and parameter regimes (Re, St)** while preserving or systematically modifying expected physical relations [1,2].
- This is exactly the idea of **physics-derived OOD transformations**: move the system out of the nominal training distribution along dimensionless-physics axes, while keeping well-defined expectations on response.
- **Constraint-rule coupled MT and mutation testing**:
 - MT combines **physics constraints (“rules”)** with empirically observed patterns to define **constraint-rule coupled metamorphic relations** [1].
 - **Mutation testing strategies** introduce perturbations to inputs and parameters, with MRs derived from **Re/St criteria** acting as oracles for whether the mutated scenario still satisfies correct scaling or similarity [2].
- **Quantitative evaluation of physical consistency and extrapolation**:
 - Both works explicitly aim to **quantify physical consistency** and **extrapolation capability** of ML models through the degree to which MRs are satisfied under perturbation/cross-domain scenarios [1,2].
 - They move beyond simple test-set errors, framing model quality in terms of **conformance to physically motivated relations under OOD conditions**.

How far they go toward operating-boundary characterization:

- **Scenario-driven multi-attribute evaluation** [1]:
 - Associates each scenario class (benchmark/extreme/perturbation/cross-domain) with **quality attributes** (functional correctness, stability, physical consistency, extrapolation, real-time performance).
 - Produces a **multi-dimensional view** of performance across scenarios—virtually a coarse **operating-characterization** over regime types.
- **Quantitative MR violation metrics in Re/St space** [2]:
 - Violations of Re/St-based MRs provide **scalar indicators** of physical inconsistency and failed extrapolation.
 - However, **they stop short of constructing explicit reliability maps** in continuous parameter space (e.g., a contour of acceptable violation rates vs Re, geometry, etc.).

Limitations relative to your full scaffold:

- **MR set is relatively narrow**: focused on **global/dimensionless correlations** (Re, St, similarity), rather than a rich multi-layer set including:
 - Local PDE residuals and weak-form constraints,
 - Boundary/interface residuals,
 - Symmetries, composition tests, etc.
- **No explicit regime (domain-of-validity) ontology per relation**:
 - Re/St relations inherently only hold in specific wake regimes (e.g., certain Re ranges), but this is **not systematically encoded as metadata for each MR**; violations are not explicitly typed as “out-of-relation-domain” vs “model failure.”
- **No formal violation-rate ‘operating-envelope’ mapping**:
 - Violations are analyzed per scenario and used for multi-attribute scoring, but **not aggregated into parametric trust regions** (e.g., subsets in Re–geometry–BC space where violation rates remain below tolerance).

Takeaway: These works [1,2] are the **closest current instantiations** of your conceptual framework in a real scientific-ML setting. They:

- Demonstrate **physics-based metamorphic testing** for scientific ML surrogates,
- Use **dimensionless, physically derived OOD transformations**, and
- Begin to address **physical consistency and extrapolation** in a systematic way.

However, they **do not yet deliver the general, relation-rich, statistically grounded operating-boundary framework** you envision. They primarily provide **one strong class of relations (dimensionless correlations)** and a **scenario-based lens** on extrapolation behavior.

Metamorphic / Relation-Based Testing in Simulation and CPS: Methodological Foundations

Generic scientific simulation and software validation via MT – Olsen & Raunak, Kanewala & Chen [6,9]

- **Problem addressed:** Validation of complex **simulation models** (ABM, DES) and generic **scientific software**, where traditional oracles are missing [6,9].
- **Contributions:**
 -

Present **MT as a systematic validation technique**, with **pseudo-oracles** defined as relations among multiple simulation runs [6,9].

- Offer **guidelines for identifying MRs**, including monotonicity, conservation-like patterns, and high-level invariants, suitable for non-testing-expert scientists [9].
- **Relevance to your goal:**
 - They **establish MT as an accepted, practical method for scientific models**, including non-differentiable and stochastic simulations.
 - Conceptually support your move from “ground-truth error checking” to **relation-based oracles**.
- **Gaps:**
 - **Little physics-specific content:** relations are mostly behavioral or monotone patterns, not PDE/similarity-based.
 - **No systematic OOD transformations** anchored in physics.
 - **No explicit violation-rate-driven operating envelopes** or regime validity modeling.

Control-theoretic design assumptions as metamorphic relations for CPS – Mandrioli et al. [4]

- **System under test:** CPS controllers interfacing with physical plants.
- **Key ideas:**
 - Replace missing oracles with **metamorphic relations derived from control-theoretical design assumptions** (stability, tracking properties, etc.) [4].
 - Use **genetic programming** to generate complex input traces from simple seeds (step/ramp), with MRs providing expected output traces and **deviation measures** guiding search [4].
- **Why this matters conceptually:**
 - Demonstrates that **design assumptions themselves can be encoded as MRs** and used as automated oracles.
 - These assumptions effectively define a **domain-of-validity**: when violated, you must distinguish:
 - Plant/input conditions outside the intended assumption domain, vs.
 - Actual controller defects.
 - This is very close to the **relation-domain validity** notion you want for PDE-based scientific ML.
- **Limitations:**
 - Focus is on CPS control, not PDE surrogates or scientific ML.
 - OOD aspects are about enriching input traces, not **physics-based similarity transformations**.

Automated MR discovery in ocean models – Hiremath et al. [5]

- **Context:** Large-scale **ocean system models** (two implementations) with complicated inputs.
- **Contribution:**
 - Proposes ML-based **automated identification of metamorphic relations**, initializing from identity maps and encouraging discovery of orthogonal, non-trivial transformations [5].
 - Uses **mutation of high-variance input attributes** and looks for relations that preserve key outputs.
- **Relevance:**
 - Addresses **MR identification in high-dimensional physical codes**, which is a critical practical challenge for your envisioned framework.
 - Suggests that **data-driven MR discovery** can complement physics-analytic MR design.
- **Limitations:**
 - Applied to traditional numerical models, not ML surrogates.
 - Physical meaning of discovered MRs (symmetries, invariances) is not systematically tied to **dimensionless relations, PDEs, or regime validity**.

Takeaway: These works collectively provide **methodological building blocks**:

- How to frame **simulation/CPS behavior as metamorphic relations** when oracles are missing.
- How to anchor MRs in **design assumptions** (a proxy for domain-of-validity).
- How to **discover MRs automatically** in complex physical software.

They **do not** address PINNs/neural operators directly, but the **assumption-based MR viewpoint** [4] and **automated MR discovery** [5] are particularly instructive for how you might:

- Encode **governing-equation, BC, and regime assumptions as explicit relation domains**, and

- Scale MR design beyond hand-written relations.

Physics-Residual-Based Verification and UQ for Neural PDE Surrogates and PINNs

Physics-informed conformal prediction (neural PDEs) – Gopakumar et al. [3]

- **Systems:** “Neural PDEs” – general neural surrogates for PDE-governed systems (e.g., plasma, heat transfer).
- **Core mechanism:**
 - Implement **convolutional layers as finite-difference stencils** to compute **PDE residuals** on model outputs [3].
 - Use residuals as **nonconformity scores** in **conformal prediction**, producing **prediction sets with coverage guarantees**—without needing labeled data [3].
- **Alignment with your interests:**
 - Treats **PDE residuals as primary physics-based diagnostics**, a strong match to your notion of **residual-based oracles**.
 - Provides **statistical coverage guarantees** on residual-based inconsistency, giving a rigorous way to **calibrate residual thresholds** and understand **how often residuals exceed tolerances** across domains.
- **Missing elements:**
 - No use of **metamorphic or OOD transformations**; test cases are ordinary PDE instances, not constructed via similarity or symmetry transformations.
 - Reliability is framed as **coverage of prediction intervals or residual-based scores**, not as **multi-relation feasible domains** over parameter/configuration space.
 - No explicit modeling of **regime-dependent domain-of-validity** for the PDE itself.

Certified neural operators and PINNs – Pasupuleti [10,11]

- **Systems:**
 - Neural operators (Fourier, DeepONet variants) [10].
 - PINNs for multiple PDE families [11].
- **Key features:**
 - Physics-informed training: residuals, BC/IC constraints, energy-consistency terms [10,11].
 - **Certification layer:**
 - **Interval bound propagation (IBP)** over input function spaces.
 - **Lipschitz constant estimation and conformal calibration** [10].
 - **Posterior residual analysis tied to collocation density and network capacity** [11].
 - Output: **rigorous instance-wise error bounds** and coverage properties, meant for regulator-ready verification [10,11].
- **Connection to your framework:**
 - Supply a principled way to relate **residual magnitudes and discretization/training choices to guaranteed solution error bounds**, which you need for:
 - Choosing **tolerances** on relation residuals, and
 - Interpreting violation magnitudes in physically meaningful scales (relative to a trusted solver).
- **Gaps:**
 - **No metamorphic testing:** residuals are used for training and certification, not as relations across transformed test cases.
 - **No explicit operating-domain maps:** guarantees are global or instance-wise, not expressed as boundaries in parameter/Re/geometry space.
 - **Domain-of-validity** of PDE models is assumed, not explicitly analyzed or tagged per relation.

Takeaway: These works are crucial for **verification and uncertainty quantification** of scientific ML surrogates, providing:

- **Physics-residual-based nonconformity measures,**
- **Error bounds and coverage guarantees,**
- And a path to **calibrate acceptable residual/violation thresholds.**

They can be seen as a **verification backbone** that your metamorphic framework could build on, even though they **do not themselves perform metamorphic/OOD relation testing or operating-envelope construction.**

Hybrid ML–Solver Frameworks with Residual-Based Trust Logic

Residual-triggered ML–CFD coupling (XRePIT) – Baral et al. [7]

- **System:** Autoregressive neural surrogate for 3D buoyancy-driven flows, coupled with OpenFOAM.
- **Mechanism:**
 - Continuously monitor **PDE residuals** of the surrogate's solution; when residual exceeds a **threshold**, control switches back to OpenFOAM [7].
 - Achieves long, stable simulations beyond standalone surrogate stability, with **wall-clock speedup and bounded error**.
- **Relevance to your operating-boundary concept:**
 - Implements a **dynamic, residual-based operating window**:
 - ML surrogate is trusted **while residuals stay below tolerance**; beyond that, solver takes over.
 - This is essentially an **online, residual-based trust policy**, akin to a (time-varying) operating envelope in state space.
- **Limitations:**
 - Residual thresholds are tuned for **runtime stability and accuracy**, not derived from systematic **relation violation statistics across a designed test suite**.
 - No metamorphic testing or physics-derived OOD transforms.
 - No explicit **regime-level domain-of-validity annotation**.

Time-marching DeepONet–FE coupling – Wang et al. [8]

- **System:** DeepONet neural operator coupled with FEM via Schwarz domain decomposition; time-marching integrated into DeepONet.
- **Mechanism:**
 - Use **error/residual measures and convergence behavior** at FE–DeepONet interfaces to decide how the ML-resolved subdomain evolves [8].
 - **Adaptive subdomain evolution** expands ML use only where accuracy is satisfactory.
- **Relation to operating domains:**
 - Similar to XRePIT, defines an **implicit “where ML is good enough” region**, but through **domain-decomposition adaptation** rather than explicit parameter-space mapping.
- **Gaps:**
 - No metamorphic or physico-OOD testing; no violation statistics-based envelope.

Takeaway: These hybrid frameworks operationalize part of what you want:

- **Residuals as actionable trust signals** for scientific ML components.
- A **dynamic “reliable operating domain”** albeit in state/space–time coordinates, used to govern coupling to a baseline solver.

However, they **lack an offline, relation-based metamorphic test layer** and **do not express reliability as a parameter/configuration-space boundary** derived from violation statistics.

What Is Still Missing and How the Pieces Fit Together

Missing elements relative to your target framework

Across all references, **no single work** yet offers:

1. A **comprehensive suite of physics-derived executable relations** for scientific ML systems that:
 - Spans **local PDE residuals, weak forms, boundary conditions, conservation laws, symmetries, similarity relations, constitutive closures, and qualitative inequalities**, and
 - Treats them explicitly as **test oracles post-training**, distinct from what went into the training loss.
1. A **systematic catalog of physically derived OOD/metamorphic transformations**, grouped into the sorts of classes you outlined (symmetry-based, parameter-scaling-based, geometry/topology perturbations, resolution/discretization perturbations, operator-composition tests), **applied to PINNs/neural operators/learned simulators** as follow-up tests.
1. A **statistical framework that converts multi-relation violation statistics into explicit reliable operating envelopes**:
 - Estimating, for each region of parameter/configuration space, the **probability and magnitude of relation violations**,

- Defining a **trusted domain** as the subset where all critical relations remain within calibrated tolerances, and
- Providing **confidence intervals** or robustness margins for these boundaries.

1. An **explicit treatment of domain-of-validity for each relation**:

- Tagging each MR (PDE residual, similarity relation, closure law) with **regime constraints, BC assumptions, discretization requirements, and implementation assumptions**,
- Classifying each observed violation as either “**out of relation domain**” or **true model inconsistency**,
- And enforcing **logical consistency** among overlapping/nested relation domains.

How current works can be synthesized

The literature you found **does provide most of the raw ingredients**:

- **Physics-based metamorphic tests for scientific ML flows** [1,2]:– Concrete examples of using **dimensionless similarity and empirical correlations (Re, St)** as MRs for ML surrogates, with **scenario-based OOD testing**.
- **Assumption- and design-based MR construction** [4] and **general MT guidance** [6,9]:– Methodological basis for deriving MRs from **governing assumptions** and for teaching non-testing experts how to systematically express such relations.
- **Automated MR discovery for physical models** [5]:– Tools to **scale MR identification** in complex PDE/simulation contexts, potentially extendable to scientific ML.
- **Residual-based nonconformity, error bounds, and conformal guarantees** [3,10,11]:– Statistical machinery to **calibrate residual/violation thresholds** and provide **coverage guarantees**; a way to connect violation magnitudes to **numerically meaningful tolerances** and **solution error bounds**.
- **Residual-threshold-based dynamic operating logic in hybrid solvers** [7,8]:– Demonstrations that **residual metrics can drive runtime trust decisions** in ML-enhanced solvers, suggesting how offline violation-rate analysis could feed into **online switching policies**.

Implications for your own work

Your project sits **one natural step beyond** the current state:

- Take the **physics-based metamorphic foundations** of [1,2] and **general MT methodologies** [4,5,6,9].
- Combine them with **residual-based UQ/certification techniques** [3,10,11] to:
 - Calibrate **physically meaningful thresholds**,
 - Quantify **violations probabilistically**, and
 - Relate relation residuals to **solver-level error baselines**.
- Incorporate **hybrid-residual trust logic** [7,8] to:
 - Use **relation violation statistics** to design **adaptive switching or refinement policies**,
 - And translate **offline operating-envelope characterization** into **online decision rules**.

Most crucially, the literature highlights that **no one has yet**:

- Constructed a **multi-relation, regime-aware validation framework** for PINNs/neural operators/learned simulators that:
 - Uses **physics-derived metamorphic OOD transformations** as first-class tests, and
 - Expresses reliability through **relation-level violation-rate maps** with **explicit domain-of-validity annotations**.

This open gap—and the partial but highly relevant examples already in the literature—both validate your problem formulation and provide **clear points of leverage** and **comparison baselines** for your eventual contributions.

Categories

Overview of How the Papers Relate to Physics-Based Metamorphic / Relation-Oracles for SciML

The papers span three fairly distinct clusters:

- **Physics-based metamorphic / relation-based testing for scientific or CPS/simulation software**– Scientific ML for physical fields: [1,2]– CPS / ocean models / general scientific software: [4,5,6,9]
- **Physics-informed uncertainty / certification for neural PDE surrogates and PINNs**– Conformal / residual-based UQ: [3]– Certified neural operators and PINNs: [10,11]
- **Hybrid ML–numerical solvers with residual-triggered coupling**– Residual-threshold coupling in CFD: [7]– DeepONet–FE time-marching and domain decomposition: [8]

Below, these are compared along dimensions that matter for your topic: type of system, nature of oracles/relations, use of physically-derived OOD/metamorphic transforms, explicit violation statistics, operating boundary characterization, and treatment of domain-of-validity.

High-Level Comparative Table

Table 1 – Alignment with Physics-Based Metamorphic / Relation-Oracles and Operating-Boundary Characterization

Ref	System under test	Metamorphic / relation-based oracles?	Physics-derived OOD / follow-up transforms?	Violation statistics & operating domain?	Domain-of-validity of relations explicated?	Direct focus on PINNs / neural operators / ML surrogates?
[1]	"Intelligent algorithms" for physical field prediction (flow around cylinder)	Yes – "constraint-rule coupled metamorphic testing" with multi-parameter physical correlations and statistical patterns; metamorphic relations used as test oracles [1].	Yes – disturbance, extreme, and cross-domain scenarios; explicitly uses multi-parameter physical correlations (e.g., similarity conditions) to generate follow-up tests [1].	Yes – "scenario-driven multi-attribute evaluation"; evaluates functional correctness, stability, physical consistency, extrapolation; interprets scenario-specific metrics but does not fully formalize a mathematical trust region [1].	Partially – scenarios are structured as benchmark/extreme/perturbation/cross-domain regimes; implicitly encode regime validity but not a formal per-relation domain ontology [1].	Yes – scientific ML surrogates for physical fields. PINN/neural-operator specificity is not emphasized, but the methodology is clearly targeted at ML-based field predictors.
[2]	Intelligent models for fluid velocity prediction	Yes – metamorphic test method integrating physical constraints; metamorphic relations derived from Reynolds number conservation and Strouhal correlations [2].	Yes – disturbance and cross-domain scenarios constructed via Reynolds/Strouhal-based transforms; these are physically derived OOD follow-up tests [2].	Partially – "quantitative evaluation of physical consistency and extrapolation capability," but focus is on scalar metrics rather than mapping violation distributions into regions of parameter space [2].	Partially – explicitly tied to regimes characterized by Re, St; relation validity implicitly constrained to flows where these correlations hold (e.g., cylinder wake), but not systematically formalized [2].	Yes – targeted at ML-based flow field surrogates; architecture-agnostic but clearly within scientific ML.
[3]	Neural PDE models (neural surrogates / "neural PDEs")	No – uses physics residuals as non-conformity scores for conformal prediction, but not framed as metamorphic relations between multiple executions [3].	No – OOD in the conformal sense (different domains) rather than physics-constructed follow-up transforms [3].	Yes – residual-based non-conformity scores and conformal coverage; defines marginal/joint coverage sets across domains [3]; but reliability is in terms of prediction intervals, not relation-violation envelopes.	Limited – requires a PDE model and a residual discretization; recognizes that physics residuals encode governing equations, but domain-of-validity is not treated regime-explicitly [3].	Yes – neural PDE surrogates; model-agnostic across architectures.
[4]	CPS controllers interacting with physical plants	Yes – defines metamorphic relations based on control-theoretic design assumptions to generate arbitrary input traces and expected outputs [4].	Partially – transforms are driven by control design assumptions rather than classical PDE symmetries/scalings; they induce systematic changes in input traces (effectively metamorphic OOD tests) [4].	Partially – deviation from expected output traces guides genetic programming search; statistics are used for test generation, not to explicitly define a reliability boundary [4].	Yes (for CPS design) – "design assumptions" explicitly delimit where the relations (guarantees) apply; violations can indicate assumption-breaking vs. controller faults [4].	No – CPS software, not scientific ML/PINNs; methodology is conceptually transferable but not instantiated on neural PDE surrogates.
[5]	Ocean system models (two implementations)	Yes – automated identification of metamorphic relations, initially represented as identity maps; uses symmetry/structure in ocean models [5].	Implicitly – metamorphic relations often correspond to symmetries of the physical system (e.g., invariances), hence define structured transforms of inputs [5].	No – focuses on identification of MRs and use for testing; does not map violation rate vs. parameter regimes into an operating envelope [5].	Limited – recognizes physical symmetries of ocean dynamics; does not formalize per-relation applicability regions [5].	No – traditional ocean models rather than ML surrogates; yet highly relevant for the metamorphic-testing-for-physics angle.
[6]		Yes – general framework to apply MT as pseudo-or	No explicit physics-based OOD transforms;	No – uses MT qualitatively to validate simulations;	Limited – discusses conceptual validity of simula-	No – not scientific ML; generic

	Simulation models (ABM, DES)	acles for simulation validation [6].	MRs are simulation-behavioral relations rather than PDE/symmetry-driven transforms [6].	does not analyze violation rates as functions of parameter space to carve operating regions [6].	tion behavior, but not formal domain-of-validity labeling per relation [6].	executable simulations.
[7]	Hybrid neural surrogate + Open-FOAM CFD solver (3D buoyancy-driven flow)	No metamorphic relations; residuals act as triggers for switching between ML and CFD [7].	No – residual thresholds are used adaptively in time, not via explicit physics-based follow-up transformations [7].	Partially – residual thresholds define when ML predictions are “trusted”; this implicitly defines a state-space-dependent operating window, but not in terms of relation-violation statistics across a parameter space [7].	Limited – thresholds tied to residuals and numerical tolerances, but domain-of-validity of governing equations not explicitly encoded [7].	Yes – ML surrogate (including finite-volume Fourier neural operator) in a hybrid solver.
[8]	DeepONet–FEM hybrid (time-marching neural operator–FE coupling)	No – uses error/residual analysis and domain decomposition, but not metamorphic or relation-based oracles [8].	No – adaptation is in spatial domain decomposition and time-marching, not through physics-based OOD transforms of test inputs [8].	Partially – evaluates error margins (<3%) and convergence rates; uses adaptive subdomain evolution as a performance criterion; but no explicit violation-rate operating envelope [8].	Limited – leverages numerical-analysis concepts (e.g., Schwarz coupling) but does not annotate domains where particular physical relations hold [8].	Yes – DeepONet neural operator as ML component.
[9]	Generic scientific software (non-ML)	Yes – advocates metamorphic testing for scientific codes; provides guidelines and examples [9].	Sometimes – examples may include invariance-like relations, but not systematically tied to physical similarity transformations [9].	No – focuses on demonstrating MT’s effectiveness, not on statistical violation envelopes or trust regions [9].	No – does not deeply engage with physical domains-of-validity; more about test design methodology [9].	No – not specific to ML.
[10]	Physics-informed neural operators with verification layer	No – uses physics residuals, energy penalties, and verified error bounds; but not metamorphic relations [10].	No – OOD not via physics-based metamorphic transforms; focuses on Lipschitz-based bounds and conformal calibration [10].	Yes – certified upper bounds and empirical coverage; but reliability is framed via error bounds/coverage rather than relation violation regions [10].	Limited – implicitly constrained by PDE type and training domain; no explicit regime-level domain-of-validity handling [10].	Yes – neural operators.
[11]	PINNs with certified error bounds	No – error certification and interval bound propagation, not metamorphic relations [11].	No – no explicit OOD transformation regime [11].	Yes – certified bounds vs. empirical L2 error; but again focused on error metrics, not relation-violation-based operating envelopes [11].	Limited – recognizes link between collocation density, residuals, and error bounds, but does not formalize per-relation domains-of-validity [11].	Yes – PINN-based neural PDE solvers.

Comparison of Metamorphic / Relation-Oracles and Physics-Based Transform Classes

Table 2 – Nature of Relations and Transformations

Ref	Type(s) of executable relations	Physics content of relations	Follow-up / OOD transformations	Relation granularity (local PDE, integral, symmetry, etc.)
[1]	“Constraint-rule coupled metamorphic relations” built from multi-parameter physical correlations and	Uses physical similarity concepts and empirical correlations (e.g., dimensionless groups) to connect scenarios; focuses on ve-	Scenario-based: benchmark, extreme, perturbation, cross-domain. Transformations include varying Reynolds-number-like parameters, boundary conditions, and perturbations	Mixed: scenario-level relations (e.g., similarity of dimensionless global quantities) rather than explicit pointwise PDE residuals;

	empirical patterns of key flow parameters [1].	locity fields around cylinder [1].	to create physically interpretable OOD cases [1].	emphasis on macroscopic flow characteristics.
[2]	Metamorphic relations based on Reynolds number conservation criterion and Strouhal number correlation [2].	Strongly grounded in classical fluid mechanics dimensionless analysis (Re, St), capturing vortex shedding properties and scaling [2].	Disturbance and cross-domain scenarios generated by altering flow conditions while preserving or systematically modifying Re/St; acts as controlled OOD transformation anchored in similarity theory [2].	Primarily integral/global and parameter-level relations (e.g., shedding frequency scaling), not local PDE residuals.
[3]	Physics residuals as non-conformity scores (e.g., PDE residuals via convolutional stencils) [3].	Directly encodes governing PDEs via discrete operators; residuals measure violation of PDE strong form [3].	No explicit transformations; different “domains” for conformal sets correspond to test regions but not metamorphic transforms of given cases [3].	Local residuals (finite-difference-like), aggregated into nonconformity metrics over spatial domain.
[4]	Metamorphic relations derived from control design assumptions; relate input-output traces via control-theoretic properties (e.g., response to arbitrary input sequences) [4].	Design assumptions encode stability, tracking, etc., which are essentially physical/plant-relational properties [4].	MRs transform baseline test inputs (step/ramp) into more complex traces while preserving certain control-theoretic guarantees; not classical PDE OOD, but physically meaningful in CPS context [4].	Time-series level (trace relations); not explicitly PDE-residual-based.
[5]	MRs represented and learned as identity maps perturbed to capture physical symmetries; orthogonality constraints ensure distinct relations [5].	Takes advantage of symmetries in ocean models (e.g., coordinate transforms, invariances) though details are domain-specific [5].	Mutations of high-variance attributes in input parameter space to identify distinct MRs; these essentially act as symmetry-/structure-based transformations [5].	Typically symmetry and structural relations at input/parameter level; less emphasis on PDE residuals.
[6]	Pseudo-oracles based on relations between model parameters and behaviors for ABM/DES [6].	Physics content minimal; more about logical/behavioral constraints (e.g., monotonicity of outputs with respect to parameters) than PDE governing laws [6].	Uses parameter perturbations as follow-ups but not systematically tied to physical dimensionless analysis [6].	Behavioral properties (monotonicities, qualitative trends).
[7]	Residual thresholds: PDE residuals of hybrid solution used to trigger fallback to CFD [7].	Strongly PDE-based: OpenFOAM residuals measure divergence from governing equations; residual threshold acts as verification metric [7].	No metamorphic transformations; but time evolution of residual under varying conditions implicitly explores different regions of state-space [7].	Local and global residual norms; no explicit relation chains.
[8]	Error/residual measures at FE–DeepONet interface; convergence metrics [8].	FE residuals/error metrics tied to PDEs; coupling ensures consistency across subdomains [8].	No; adaptation is spatial/time decomposition rather than transformed inputs.	Residuals and convergence properties at domain-decomposition-level.
[9]	General MRs (e.g., invariance, monotonicity, conservation-like properties) for scientific programs [9].	Highly generic; can encode physical laws when applied to physics codes, but the paper mostly illustrates generic scientific computation cases [9].	Variable; in principle may include physically-derived transformations, but not systematized for PDE similarity or dimensionless groups [9].	Mixed – from simple scalar relations to more complex functional/invariance relations.
[10]	PDE residuals, energy-consistency constraints, boundary/IC conditions; Lipschitz and IBP-based error bounds [10].	Directly encode PDEs and energy principles; error bounds are mathematical rather than physical-similarity relations [10].	No OOD metamorphic transforms, though coverage under distribution shifts may be studied through conformal calibration [10].	Local residuals + global energy norms; error-bounds at solution-operator level.
[11]	Collocation residuals, IBP-based error bounds; posterior residual analysis [11].	PDE residuals are central; error certification uses them plus interval-analysis machinery [11].	No; focuses on certification on given test sets, not metamorphic OOD transformations.	Local residuals aggregated into global L2 error bounds.

Treatment of Violation Statistics, Reliability, and Domain-of-Validity

Table 3 – From Residuals to Reliability: How Far Each Work Goes

Ref	How violations / residuals are quantified	How reliability / trust is expressed	Treatment of domain-of-validity and regime distinctions
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[1]	Scenario-wise evaluation of performance under four categories: benchmark, extreme, perturbation, cross-domain [1]. Uses multiple metrics (functional correctness, stability, physical consistency, extrapolation) across scenarios; violations of metamorphic relations and constraints are part of this evaluation.	Constructs a “scenario-driven multi-attribute evaluation” that maps scenario characteristics to quality attributes [1]. This is closest to a multi-dimensional operating-characterization , though it is not explicitly codified as a formal feasible region in parameter space.	Scenario categories effectively encode regimes (normal vs extreme vs cross-domain). The domain-of-validity is implicitly scenario-based; there is no per-relation regime ontology, but the framework distinguishes verification vs extrapolation vs cross-domain testing [1].
[2]	Quantitative measures of physical consistency and extrapolation via fidelity to Reynolds and Strouhal-based relations (e.g., correctness of shedding-frequency correlations under disturbed/cross-domain cases) [2].	Uses violation magnitudes of Re/St-based MRs to evaluate extrapolation capability and physical consistency [2]; reliability is still reported as scalar/aggregate metrics, not as a mapped boundary in Re–parameter space.	Re and St criteria inherently restrict relations to certain flow regimes (e.g., cylinder wakes under moderate Re); this is acknowledged through their use but not systematically expressed as explicit domain filters or labels [2].
[3]	Physics residuals (computed via convolutional finite-difference stencils) define nonconformity scores; distributions of these scores are used in conformal prediction [3].	Reliability is expressed as prediction sets (conformal intervals/regions) with marginal and joint coverage guarantees over prediction domains [3]. These intervals reflect residual-based inconsistency with the PDE, not only data error.	Assumes governing PDE is valid across considered domains; no explicit separation between regions where PDE model is itself invalid vs where ML approximation fails.
[4]	Deviation between actual and expected output traces (derived from MRs) guides genetic programming for input generation [4]. Violations are measured as trace deviations.	Reliability is implicit: inputs for which deviations are large reveal violations of design assumptions and possible CPS defects; the main goal is fault detection, not explicit operating-envelopes [4].	Design assumptions explicitly delineate when control-theoretic guarantees hold; violations may be interpreted as assumption break or implementation failure [4]. This is one of the clearest treatments of assumption-domain validity , albeit in CPS rather than PDE SciML.
[5]	Uses cost function minimizing distance from identity map and enforcing orthogonality between MRs; violations are differences between outputs under MR-induced input mutations [5].	Reliability understood as ability to distinguish two implementations via MR violation patterns; there is no explicit operating domain map [5].	Focuses on discovering MRs; does not explicitly encode regimes; domain-of-validity remains implicit in the chosen input attribute ranges.
[6]	MT violations are detected as failures of expected relations between parameters and simulation outcomes in ABM/DES case studies [6].	Simulation validity is argued via presence/absence of systematic MT violations [6]; not mapped to parameter-space boundaries.	Not explicitly considered; relations are assumed valid throughout the studied parameter ranges.
[7]	Monitors PDE residuals of hybrid solution; residual thresholds determine when to hand back to OpenFOAM [7]. Relative L2 errors also reported.	Reliability is effectively defined as “state-time regions where the residual is below threshold,” which act as operating windows for surrogate-only evolution ; outside, the solver is engaged [7]. This is a <i>dynamic</i> operating envelope, but not parameter-space mapped.	Thresholds are set in context of solver tolerances; domain-of-validity of the governing equations is inherited from OpenFOAM settings but not encoded per relation [7].
[8]	Error margins (<3%), convergence rates, and residuals at FE–DeepONet interfaces [8].	Reliability = meeting error margin targets and convergence properties; the adaptive subdomain evolution ensures that ML is used primarily where it remains accurate [8].	Assumes validity of PDEs and FE formulations; does not discuss regime-dependent validity of physical laws or relations.
[9]	MT violations measured case by case; mostly illustrative examples [9].	Reliability is improved via MT-guided bug detection; no notion of operating boundary [9].	Domain-of-validity not developed; focus is methodological MT guidance.
[10]	L2 errors, energy-norm errors, certified upper bounds via Lipschitz/IBP, empirical coverage vs target levels [10].	Reliability expressed as certified error bounds and coverage probabilities over input-function space [10]; these bounds act as “guaranteed envelopes” but are not tied explicitly to multiple physical relations.	Relations are PDEs and energy balances assumed valid over the training/testing domain; no explicit handling of regime-specific applicability or BC-dependent validity.
[11]	Certified L2 error bounds vs empirical error; residual-based posterior analysis [11].	Reliability defined through closeness of error bounds to actual L2 error and regulator-ready reporting; essentially a verification perspective [11].	PINN assumptions (collocation density, PDE validity) are global; no explicit regime or domain-of-validity partitioning.

Focused Comparison: How Close Each Work Gets to Your Target Concept

Your target concept is:

- Physics-derived **metamorphic / property-based oracles** for scientific ML (PINNs, neural operators, learned simulators).
- Follow-up **physically derived OOD transformations** (symmetries, Re/Ma scaling, geometry perturbations, etc.) as **tests**.
- Use of **relation-level violation rates** to **characterize a reliable operating boundary**, with explicit acknowledgement of **relation domain-of-validity**.

Closest matches in spirit

- [1] Qi et al. (multi-scenario constraint-rule metamorphic testing for physical field prediction)
 - **Scientific ML context**: explicitly targets “intelligent algorithms” for physical field prediction (CFD-like) [1].
 - **Metamorphic & physics-based**: uses constraint-rule coupled metamorphic relations built from multi-parameter physical correlations and empirical flow patterns [1].
 - **Physically derived OOD scenarios**: perturbation and cross-domain scenarios intentionally push the model beyond nominal training conditions while maintaining physically interpretable relations [1].
 - **Multi-dimensional operating characterization**: scenario-driven multi-attribute evaluation approximates a multi-axis operating envelope (functionality, stability, physical consistency, extrapolation) [1].
 - **Domain-of-validity**: encoded at scenario level (benchmark vs extreme vs cross-domain), but not explicitly per relation.
- [2] Yu et al. (Reynolds/Strouhal-based metamorphic testing for fluid velocity prediction)
 - **Scientific ML context**: fluid-velocity ML predictors (similar to surrogates for CFD) [2].
 - **Metamorphic & physics-based**: relations grounded in Reynolds number conservation and Strouhal correlations; strongly aligned with dimensionless similarity theory [2].
 - **Physically derived OOD**: disturbance and cross-domain scenarios explicitly generated by altering Re/St regimes [2].
 - **Relation violations as extrapolation metrics**: uses violation magnitudes of these correlations as quantitative measures of physical consistency and extrapolation capability [2].
 - **Domain-of-validity**: implicitly constrained to flow regimes where classical Re/St relationships are valid; however, this is not formally encoded as per-relation domain filters.

These two papers are the **closest direct instantiations** of the conceptual scaffold you described, though they do not yet:

- Use **rich sets of local/weak-form PDE residuals, symmetries, and composition tests** in addition to dimensionless correlations.
- Explicitly turn violation statistics into a **formal operating-envelope** in parameter/state space.
- Provide a **formal ontology** of relation domains-of-validity.

Conceptually complementary but not fully aligned

- [4] Mandrioli et al. – control-theoretic design assumptions as metamorphic relations for CPS.
 - Strong treatment of **assumption-domain validity** and **metamorphic relations**; could inform how to encode relation domains-of-validity for scientific ML.
 - Does not address PINNs/neural operators, nor physics-derived OOD via PDE similarity.
- [5] Hiremath et al. – automatic MR discovery for ocean models.
 - Useful for **MR identification** and handling high-dimensional physics models; but not ML-specific, nor focused on operating envelopes.
- [6,9] Olsen & Raunak; Kanewala & Chen – general MT for simulation/scientific software.
 - Provide overall **MT frameworks** but lack the **physics-specific OOD and regime-validity** elements you require.

Residual-based reliability / certification but not metamorphic

- [3] Gopakumar et al. – physics-based conformal prediction for neural PDEs.
 - Uses **physics residuals as nonconformity scores**, enabling **data-free UQ** with **coverage guarantees** [3].
 - Reliability is expressed via **prediction sets** rather than multi-relation violation envelopes.
 - No metamorphic OOD transformations or explicit regime-wise validity.
- [10] Pasupuleti (neural operators) and [11] Pasupuleti (PINNs)
 - Provide **certified error bounds** with interval bound propagation and residual analysis [10,11].
 - Very strong on **verification** and **instance-wise guarantees**, but again not using metamorphic relations nor domain-wise violation-rate envelopes.

These works could be combined with metamorphic testing frameworks to provide **certified residual/error thresholds** that define boundaries of trust regions, but they do not themselves implement the metamorphic / OOD-relational testing layer.

Hybrid coupling and residual-based trust regions

- [7] XRePIT and [8] Time-marching neural operator–FE coupling
 - Use **residual thresholds and error margins** as triggers for switching between ML and traditional solvers [7,8].
 - This operationalizes a **dynamic operating boundary**: ML is trusted while residuals are small, and the solver takes over when they exceed thresholds.
 - However, the boundary is **implicit and time-/state-dependent**, and not expressed as a **map over physical parameter space** or as violation-rate statistics of multiple relations.
 - No metamorphic follow-up tests or explicit regime-validity treatment.

These hybrid frameworks are **key examples of residual-based operational trust strategies** that could be extended with metamorphic relation testing and regime-aware domains.

Synthesis: What Is Unique and Where the Gaps Are

Unique contributions relevant to your topic:

- [1,2] are the only works in this set that:
 - Explicitly implement **metamorphic testing for scientific ML models** in fluid dynamics.
 - Construct metamorphic relations directly from **dimensionless physical correlations** (Re, St) and **multi-parameter physical similarity** [1,2].
 - Use these relations under **physically derived OOD scenarios** to evaluate extrapolation and physical consistency.
- [4] brings a **rigorous treatment of design assumptions as metamorphic relations** and clearly separates **assumption-domain validity** from system failure in CPS. That conceptual structure is directly transferable to relation-domain-of-validity for scientific ML.
- [3,10,11] provide **physics-residual-based UQ/certification** for neural PDE surrogates and PINNs, a crucial ingredient if you want to calibrate **tolerances** for relation violations in a physically meaningful way.
- [7,8] instantiate **residual-based operating-window control** in hybrid ML–solver loops, giving concrete examples of how residual thresholds can shape **trusted ML regions** in practice.

Key gaps with respect to your full target framework:

- None of the papers fully integrate:
 - **Rich sets of physics-based metamorphic relations** (PDE residuals, weak-form integrals, symmetries, similarity, constitutive relations)
 - With **physically constructed OOD transformations**,
 - And with **statistical characterization of violation rates across parameter/state space**,
 - While explicitly encoding **per-relation domain-of-validity** and distinguishing **out-of-model-domain behavior** from genuine model failures.
- **Relation domains-of-validity** are only partially addressed:
 - Scenario-wise / regime-wise in [1,2],
 - Explicitly through design assumptions in CPS context in [4],
 - Implicit through PDE assumptions in [3,7,8,10,11]. No work offers a systematic regime-tagging of each relation with conditions on Re/Ma, BCs, discretization, etc., in the way your scaffold suggests.
- **Reliable operating boundaries** are:
 - Implicit/dynamic (residual thresholds in [7,8]),
 - Expressed as **prediction sets / error bounds** ([3,10,11]) rather than *multi-relation feasible domains*,
 - Or expressed qualitatively via scenario-based performance in [1,2] but not as an explicit feasible-region approximation in parameter space.

These comparisons should help you position your own work: combining the **physics-based metamorphic/OOD relation design** of [1,2], the **assumption-domain discipline** of [4], the **residual-based certification** of [3,10,11], and the **operational trust-switching** of [7,8] into a unified framework for scientific ML validation that explicitly maps **relation violation statistics** into **regime-aware operating boundaries**.

Timeline

Timeline Overview and Major Phases

2019–2021: General-purpose **metamorphic testing (MT)** for scientific software and simulation is articulated and systematized, but with little focus on ML-based PDE surrogates or scientific ML. **2021–2024:** MT is adapted to **domain-specific physical models and CPS**, with increased emphasis on design assumptions and partially on automated MR discovery; this begins to look closer to physics-based V&V. **2023–2025:** In parallel, **physics-informed UQ and certification** for neural PDE surrogates and PINNs develops, providing error-bound and coverage-oriented reliability notions. **2024–2025:** Dedicated work emerges on **metamorphic and mutation testing for intelligent physical field prediction models**—directly targeting ML surrogates for fluid dynamics and introducing physically derived MRs, multi-scenario testing, and extrapolation assessment [1,2]. Hybrid ML–solver coupling frameworks use residual thresholds as operating-regime triggers [7,8], implicitly approximating operating envelopes.

The field is thus moving from **generic MT concepts for simulation** 'control-theory/physics-informed MRs in CPS and numerical simulations' **scientific-ML-specific validation frameworks with physics-based MRs, hybrid coupling, and certified UQ.**

Early Foundations: Metamorphic Testing for Scientific Software & Simulation (2019–2021)

2019: Metamorphic testing explicitly framed for scientific software

- **Kanewala & Chen (2019) [9]**
 - Audience: scientific software developers (non-professional testers).
 - Contribution: Presents **metamorphic testing as a general approach for scientific software** with oracle problems.
 - Emphasis:
 - Identification and specification of metamorphic relations (MRs) for scientific programs.
 - Practical guidance for applying MT where ground-truth outputs are unavailable.
 - Relevance:
 - Provides the **conceptual and methodological baseline** for using MRs as oracles in scientific computing.
 - However, it does *not* focus on learned PDE surrogates or systematic OOD physical transformations; relations are largely domain-agnostic patterns.
- **Olsen & Raunak (2019) – *Increasing Validity of Simulation Models Through Metamorphic Testing* [6]**
 - Focuses on **simulation validation** (ABM and DES) using MT.
 - Key ideas:
 - MT is explicitly used for **simulation validation**, not just software correctness.
 - Defines **pseudo-oracles** based on relations between parameters and behaviors in executable models.
 - Case studies show MT detecting subtle defects in complex simulations without explicit oracles.
 - Historical significance:
 - Bridges MT with **simulation V&V** directly.
 - Demonstrates that validation can be structured around **multi-run relations and behavior patterns**, a conceptual ancestor of relation-based operating-domain analysis.
 - Limitations w.r.t. your topic:
 - No explicit physics PDE context or ML surrogates; relations are simulation-level rather than PDE/physics-operator relations.
 - No explicit mapping of violation rates to reliable operating envelopes.

2021: Automated MR identification in ocean system models

- **Hiremath et al. (2021) – *Towards Automated Metamorphic Test Identification for Ocean System Models* [5]**
 - Domain: **ocean system modeling**; large input spaces; multiple implementations.
 - Contribution:
 - Proposes **automated MR discovery using ML**, representing MRs as transformations and learning non-trivial relations orthogonal to identity.
 - Uses dimensionality reduction and mutation on high-variance attributes to discover additional MRs.
 - Significance:
 - Moves MT from purely human-crafted MRs towards **data-driven MR discovery** for complex physical codes.
 - Explicitly targets a **physical modeling domain** (ocean models), where symmetries and invariances are physically meaningful.
 - Relevance:

- Early step towards **systematic, possibly automated MR design for physical systems**.
- Still focused on classical simulation implementations; does not yet target ML surrogates or operator-learning architectures.

Summary for this phase: By ~2021, the community had established MT as a **practical strategy for simulation and scientific software validation**, with some attention to physical systems and, importantly, beginnings of automated MR discovery. However, **ML-based PDE surrogates, PINNs, and neural operators are not yet central**. Relations are primarily simulation-level and not systematically tied to PDE residuals, conservation laws, or scientifically grounded OOD transformations with operating-domain characterization.

Extending Metamorphic and Relation-Based Testing to CPS and Control (2024)

- **Mandrioli et al. (2024)** – *Testing CPS With Design Assumptions-Based Metamorphic Relations and Genetic Programming* [4]
 - Domain: **Cyber-Physical Systems (CPS)**.
 - Core idea:
 - Use **control-theoretical design assumptions** as the foundation for **metamorphic relations**.
 - Generate tests through **genetic programming**, producing rich input traces beyond simple steps/ramps.
 - Deviation between actual and expected physical traces (as predicted by design-assumption-based MRs) guides test generation.
 - Contributions:
 - Reframes the oracle problem in CPS testing by leveraging **design assumptions as executable relations**.
 - Demonstrates **MRs derived from control theory** (e.g., stability, tracking behavior) with automated generation.
 - Significance for the trajectory:
 - First clear demonstration that **engineering design assumptions** (analogous to domain-of-validity conditions in physics) can **serve as metamorphic oracles** in complex CPS scenarios.
 - Provides a **pattern for using physics/control assumptions as test oracles**, aligning with your notion of relation-domain validity.
 - Limitations vs. your goal:
 - System under test is CPS software, not scientific ML solvers; no explicit neural surrogates or PINNs.
 - MRs are based on control design rather than PDE-based or operator-based transformations, and there is no explicit notion of OOD transformation in the sense of similarity transforms or dimensionless scaling.

Trend: This work exemplifies a **shift from generic MT (2019–2021) to domain-specific, assumption-based metamorphic testing** in engineered systems. It highlights that **design assumptions = domain validity conditions** can be formalized as MRs—an idea that is conceptually crucial for your desired framework.

Emergence of Physics-Informed Certification and UQ for Scientific ML (2023–2025)

Parallel to MT/V&V developments, a **certification/UQ line of work** emerges for neural PDE surrogates and PINNs. These works emphasize **physics-informed nonconformity scores, error bounds, and coverage guarantees**, which are highly relevant to defining operating domains, even if they do not rely on metamorphic relations.

2023: Certification for neural operators and PINNs

- **Pasupuleti (2023)** – *Physics-Informed AI in Python: Neural Operators with Verified Error Bounds* [10]
 - Systems: Fourier/DeepONet-style neural operators for PDEs.
 - Key elements:
 - Training with **physics residuals, energy-consistency penalties, and BC constraints**.
 - A **verification layer** using:
 - Global Lipschitz constant estimates.
 - **Interval bound propagation (IBP)** over function spaces.
 - **Conformal calibration** for distribution-free coverage.
 - Importance:
 - Provides **instance-wise certified error bounds** and empirical coverage close to target levels.
 - Conceptually introduces the idea of “**verified**” **neural operators**, which is tightly aligned with scientific-ML V&V concerns.

- Relation to your theme:
 - Uses **physics residuals as a verification metric** but not as metamorphic relations between multiple test executions.
 - The operating reliability is framed via **error bounds and conformal coverage**, not via relation-violation rates across physically transformed tests.
- **Pasupuleti (2023)** – *Quantum-Inspired PINNs in MATLAB: Certified Neural PDE Solvers with Error Bound* [11]
 - Systems: PINNs with QAOA-inspired optimization.
 - Contributions:
 - Integrates **certification of solution error** via IBP and **posterior residual analysis**.
 - Formalizes links between **collocation density, network capacity, optimizer schedules, and certified bounds**.
 - Significance:
 - Extends certification ideas to PINNs, showing how error bounds can be integrated in a verification-ready PINN framework.
 - Relevance:
 - Again, **verification via residual-based and IBP-based bounds**, not metamorphic or relation-based testing per se.
 - Still important as part of the emerging “**certified scientific ML**” trend.

2025: Physics-informed conformal prediction for neural PDEs

- **Gopakumar et al. (2025)** – *Calibrated Physics-Informed Uncertainty Quantification* [3]
 - Focus: **Model-agnostic, physics-informed conformal prediction** for neural PDEs.
 - Key techniques:
 - Convolutional layers used as **finite-difference stencils** to compute PDE residuals.
 - **Physics residuals as nonconformity scores** in conformal prediction.
 - Provides **data-free UQ and coverage guarantees** across prediction domains.
 - Significance:
 - Fully embraces **residual-based physics nonconformity** as the basis for UQ.
 - Moves beyond pointwise residual checks to **distributional guarantees (coverage) on residual-derived scores**.
 - Relevance to operating envelopes:
 - Conceptually close to **relation-level violation statistics**, but focused on **coverage probabilities of residual-based scores** rather than explicit metamorphic relations or physically constructed OOD transformations.
 - Provides a **statistical framing** for “how often do physics residuals exceed a threshold?”—a building block for relation-violation rate analysis.

Summary for this phase: From ~2023 onward, a **certification and UQ-oriented thread** emerges in scientific ML. It:

- Uses **physics residuals and energy constraints** as verification tests.
- Develops **formal error bounds and conformal coverage guarantees**.
- Contributes pieces of what you want (residual-based nonconformity, coverage across domains), but does not yet integrate **metamorphic testing, physically derived transformations, or explicit domain-validity formalization**.

Hybrid ML–Solver Frameworks with Residual-Triggered Coupling (2025)

While not directly MT, hybrid frameworks **implicitly use residual thresholds as operational criteria**, which is closely related to **operating boundary characterization**.

- **Baral et al. (2025)** – *XRePIT: Deep learning–CFD hybrid framework in OpenFOAM* [7]
 - System: Autoregressive neural surrogates + OpenFOAM CFD.
 - Key innovation:
 - **Residual-guided hybrid loop**: monitor residuals; when they exceed a threshold, fall back to CFD solver.
 - Achieves long-term stable simulations that extend beyond the surrogate’s stability horizon.
 - Relevance:
 - Uses **physics residuals as an online monitoring oracle** to decide when the ML model remains trustworthy.
 - This is a *proto-operating-domain* notion: the region where residuals stay below threshold is effectively the trusted domain.

- Limitations:
 - Does not cast this as metamorphic testing or relation-level violation rates across systematically constructed tests.
 - Residual thresholds are used online rather than for offline characterization of reliable operating boundaries.
- **Wang et al. (2025)** – *Time-marching neural operator–FE coupling* [8]
 - System: DeepONet + FEM via domain decomposition with Schwarz coupling.
 - Contributions:
 - Embeds time-stepping in the DeepONet.
 - Uses **adaptive subdomain evolution**; ML-resolved regions grow where accuracy is acceptable.
 - Relevance:
 - Again, **error/residual-driven adaptation**; some notion of where the neural operator is “accurate enough” to be used.
 - Similar to XRePIT, this suggests **implicit operating envelopes**, though not formalized via relation-based testing.

Trend: These works indicate an emerging **operational coupling paradigm**: ML surrogates are used **within bounds** defined by residual or error thresholds, with fallback to trusted solvers. This is not yet framed as metamorphic relation testing or violation-rate mapping, but it indicates recognition that **scientific ML systems must operate within empirically verified reliability regions**.

Domain-Specific Metamorphic and Mutation Testing for Intelligent Physical Field Prediction (2025)

The most directly relevant cluster for your topic appears in **Qi’s group** (and collaborators), focusing explicitly on intelligent fluid field prediction with **physics-based MRs and OOD scenarios**.

2025: Multi-scenario metamorphic testing for intelligent physical field prediction

- **Qi et al. (2025)** – *Research on Multi-scenario Collaborative Testing Methods for Intelligent Algorithms in Physical Field Prediction* [1]
 - Domain: **Intelligent algorithms for physical field prediction** (velocity field around a cylinder).
 - Core contributions:
 - A **hierarchical verification architecture** with four scenario types:
 - Benchmark, extreme, perturbation, and cross-domain scenarios.
 - Each scenario type targets different attributes: functional correctness, stability, physical consistency, extrapolation capability, real-time performance.
 - **Constraint-rule coupled metamorphic testing**:
 - Constructs metamorphic relations using **multi-parameter physical correlations** and **experimental statistical patterns** (e.g., known correlations among Reynolds number, Strouhal number, vortex shedding behavior).
 - Generates **physically interpretable test inputs** to verify physical similarity and extrapolation in critical and cross-operational conditions.
 - **Scenario-driven multi-attribute evaluation**:
 - Integrates quality attributes (e.g., stability, accuracy, physical consistency) with scenario adaptation strategies.
 - Provides a **quantifiable evaluation system**, transitioning from code-functional verification to **physical-consistency verification**.
 - Significance:
 - Directly aligns with your goal of **metamorphic testing grounded in physical relations** for scientific ML systems.
 - Explicitly addresses **extrapolation (cross-domain) testing using physically motivated scenarios**.
 - Moves beyond one-off residual checks toward systematic scenario classes, each with distinct metamorphic objectives.
 - Limitations relative to your full vision:
 - Abstract does not explicitly mention **relation-level violation distributions or explicit operating-boundary mapping**; it focuses on scenario categories and multi-attribute evaluations.
 - Domain-of-validity considerations are implicit via scenario design and physical correlation-based MRs; explicit formalization of relation domains (e.g., specific Re ranges) is not detailed in the abstract.

2025: Mutation and metamorphic testing using dimensionless physical criteria

- **Yu et al. (2025)** – *Research on Mutation Testing Strategies for Intelligent Models Predicting Fluid Velocity Fields* [2]

- Domain: **Intelligent models for fluid velocity fields** (likely similar or related setup to [1]).
- Contributions:
 - Proposes a **metamorphosis test method with physical constraints**, combining:
 - **Disturbance scenarios** and **cross-domain scenarios**.
 - **Metamorphic relations constructed from Reynolds number conservation and Strouhal number correlations**.
 - These MRs:
 - Break the bottleneck of output uncertainty by providing physically grounded expected relationships even under OOD conditions.
 - Enable **quantitative evaluation of physical consistency and extrapolation capability** of the model.
 - Emphasis on **scene scalability** for engineering deployment.
- Significance:
 - Strongly embodies **physically derived metamorphic relations** using dimensionless numbers and similarity laws, matching your desideratum of OOD transformations with known physical relations.
 - Demonstrates **mutation/disturbance scenarios** as follow-up tests to probe physical consistency beyond training distributions.
- Relation to your framework:
 - Offers **concrete examples** of physically meaningful OOD transformations (Re and Strouhal-based) and associated MRs.
 - Moves toward quantitative, physically interpretable extrapolation assessment, though again without explicit mapping of violation-rate contours in parameter space.

Cluster characteristics and impact

- **Cluster:**
 - **Lianzhi Qi** appears as a key figure spanning [1,2].
 - The work forms a **coherent research line** on **testing intelligent fluid-dynamics surrogates** using **physics-based metamorphic and mutation tests**.
- **State-of-the-art contribution:**
 - Among the listed references, [1,2] are **closest to integrating MT, physical correlations, OOD testing, and scientific-ML evaluation**.
 - They explicitly:
 - Use **physics-derived relations** (dimensional similarity, Re and St) as MRs.
 - Construct **scenario taxonomies** to probe extrapolation and cross-domain behavior.
 - They do not yet fully implement **relation-domain validity modeling** or **statistical operating-boundary mapping** as in your target vision, but they provide a **strong foundation and real-world fluid dynamics case study**.

Cross-Cutting Trends, Patterns, and Future Trajectories

Trend 1: From generic MT to domain-specific, physics-informed relations

- Early work [9,6,5] provides **generic MT frameworks** for scientific software and simulation.
- CPS-focused work [4] and fluid-dynamics-focused work [1,2] show **increasingly domain-specific MRs**:
 - Derived from **control theory and design assumptions** in CPS [4].
 - Derived from **dimensionless numbers, similarity relations, and empirical correlations** in fluid mechanics [1,2].
- Significance:
 - This evolution reflects a **shift from software-centric to physics-centric MRs**, aligning with your emphasis on **physics-based oracles**.

Trend 2: Emergence of physics-residual-based verification and UQ

- PINN and neural-operator certification works [10,11] and physics-informed CP [3] build **verification and UQ around physics residuals**:
 - Residuals as the primary **nonconformity scores**, used for error bounds or conformal guarantees.

- This aligns with the idea of “**physics residuals as relations**” and provides a statistical machinery for **violation frequency/coverage**.
- Connection to your goals:
 - These works supply **formal, statistically grounded reliability metrics** that can be combined with MT-based relation violations to define operating envelopes.

Trend 3: Hybrid ML–solver coupling with residual- or error-based regime switching

- Hybrid frameworks [7,8] use **residual thresholds or accuracy-based subdomain evolution** to decide when ML surrogates are trustworthy.
- Interpretation:
 - This is effectively **runtime operating-boundary enforcement**: stay in ML mode while residuals are acceptable; revert to high-fidelity solver otherwise.
- Potential future integration:
 - Combine such **residual-triggered switching** with **offline metamorphic and OOD testing** to design explicit **trust-region policies**.

Trend 4: Limited yet growing attention to operating-domain characterization

- Some ingredients exist:
 - **Coverage guarantees** for residual-based nonconformity [3].
 - **Scenario-based extrapolation assessments** [1,2].
 - **Residual thresholds for switching** [7].
- However:
 - No paper yet **fully maps relation-level violation rates over parameter space** as an explicit **reliable operating envelope**, while also encoding **relation-specific domain validity**.
- This gap points directly to the **future research opportunity** matching your stated goal.

Key contributors and likely future directions

- **MT and simulation validation pioneers**:
 - **Olsen & Raunak** [6], **Kanewala & Chen** [9] – established MT as a general scientific software testing approach.
 - Expect continued influence in **methodological frameworks and educational efforts** for scientists adopting MT.
- **Automated MR discovery for physical models**:
 - **Hiremath et al.** [5] – early work on ML-based MR discovery.
 - Suggests a future direction of **automated or semi-automated generation of physically plausible MRs**, potentially extended to scientific ML surrogates.
- **CPS control-theory-based MT**:
 - **Mandrioli & Briand et al.** [4] – demonstrate **design assumptions as MRs**; this could inspire similar approaches in **PDE-constrained ML** using **governing-equation assumptions and modeling regimes** as structured test relations.
- **Scientific-ML certification**:
 - **Pasupuleti** [10,11] and **Gopakumar et al.** [3] – central in **physics-informed UQ and certification**, providing the current best-developed **error-bounding frameworks**; likely to push towards **more integrated V&V pipelines with certified operators/PINNs**.
- **Physics-based MT for intelligent fluid prediction**:
 - **Qi, Yu, and coauthors** [1,2] – currently the **most directly relevant cluster** for your focus on **metamorphic testing of scientific ML for PDEs**.
 - Their progressive work suggests future directions:
 - Extending MR-based testing to more complex geometries and multiphysics problems.
 - Incorporating **statistical violation-rate analyses** and explicit **domain-of-validity annotation** for each MR.
 - Possibly integrating with **hybrid ML–solver frameworks and UQ methods**.

Overall Trajectory Relative to Your Target Vision

Across these works, the field has evolved from:

1. **Generic MT for scientific software and simulations** [9,6] '
1. **Domain-specific, assumption-driven MRs for CPS and ocean models** [5,4] '
1. **Physics-informed UQ and certified neural PDE surrogates** [10,11,3] '
1. **Physics-based metamorphic and mutation testing tailored to intelligent physical field predictors** [1,2], alongside **residual-driven operating decisions in hybrid frameworks** [7,8].

Elements of your desired framework are present in fragments:

- **Metamorphic / relation-based oracles grounded in physics** – strongly in [1,2], conceptually in [4,5,6,9].
- **Physically derived OOD transformations** – explicitly via Re/Strouhal and scenario design in [1,2]; implicitly through perturbations and symmetries in [5].
- **Residual-based nonconformity and coverage** – [3,10,11].
- **Operational boundaries via residual thresholds** – [7,8].

What is still missing (and thus points to future research):

- **Systematic use of executable relation-level violation rates** (across multiple MRs) to construct **explicit, parameter-space operating envelopes**.
- **Explicit encoding of domain validity for each relation**, distinguishing “out-of-model-domain” from “true model failure” in a structured way.
- A **unified framework** that ties together:
 - Physics-based metamorphic OOD tests,
 - Residual- and relation-based nonconformity scores,
 - Statistical violation-rate estimation and coverage, and
 - Hybrid coupling policies or control decisions based on those metrics.

The trajectory of the literature suggests that such an integrated framework is a **natural next step**—and your project is well-positioned to synthesize these strands into a coherent, V&V-grounded methodology for broad scientific ML systems (PINNs, neural operators, surrogate PDE models, and learned simulators).

Foundational Work

Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

Ref.	Reference Rate	Title	Cited By These Relevant Papers
[141]	0.52	Metamorphic Testing: A New Approach for Generating Next Test Cases	[1, 2, 5, 6]
[142]	0.46	Out-of-distribution generalization of deep-learning surrogates for 2D PDE-generated dynamics in the small-data regime	[16]
[143]	0.42	Learning Mesh-Based Simulation with Graph Networks	[1, 3]
[144]	0.27	Testing and Validating Machine Learning Classifiers by Metamorphic Testing	[1, 6]
[100]	0.25	Uncovering near-wall blood flow from sparse data with physics-informed neural networks	[1]
[105]	0.23	Physics-Informed Neural Networks for Multiphysics Data Assimilation with Application to Subsurface Transport	[1]
[111]	0.23	Physics-Informed Neural Networks for Nonhomogeneous Material Identification in Elasticity Imaging	[1]
[145]	0.22	Metamorphic testing of programs on partial differential equations: a case study	[1, 17]
[101]	0.21	Physics-informed machine learning	[3, 14, 16]
[18]	0.21	Uncertainty quantification of surrogate models using conformal prediction	[3]

[146]	0.21	Embedding machine-learnt sub-grid variability improves climate model precipitation patterns	[3]
[147]	0.21	Validating a Deep Learning Framework by Metamorphic Testing	[1]
[148]	0.21	Identifying implementation bugs in machine learning based image classifiers using metamorphic testing	[1]
[149]	0.21	The evaluation of program-based software test data adequacy criteria	[1]
[35]	0.21	Scaling physics-informed hard constraints with mixture-of-experts	[3]
[74]	0.21	Calibrated Uncertainty Quantification for Operator Learning via Conformal Prediction	[3]
[106]	0.21	Estimating the carbon footprint of computational fluid dynamics	[3]
[150]	0.21	Plasma surrogate modelling using Fourier neural operators	[3]
[151]	0.21	Active Learning for Neural PDE Solvers	[3]
[152]	0.21	Data efficiency and long term prediction capabilities for neural operator surrogate models of core and edge plasma codes	[3]

Adjacent Work

Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

Ref.	Adjacency score	Title	References These Foundational Papers
[133]	0.93	Physics-Informed Latent Neural Operator for Real-time Predictions of Complex Physical Systems	[81, 109, 118]
[34]	0.49	An RBC-MsUQ Framework for Red Blood Cell Morpho-Mechanics	[42, 81, 99]
[86]	0.41	IB-UQ: Information bottleneck based uncertainty quantification for neural function regression and neural operator learning	[42, 81, 101, 159]
[60]	0.36	Evaluating Uncertainty Quantification approaches for Neural PDEs in scientific applications	[81, 99]
[164]	0.33	Enabling Automatic Differentiation with Mollified Graph Neural Operators	[35, 123]
[165]	0.33	Towards Enforcing Hard Physics Constraints in Operator Learning Frameworks	[35, 123]
[81]	0.32	NeuralUQ: A comprehensive library for uncertainty quantification in neural differential equations and operators	[42, 99, 115]
[35]	0.31	Scaling physics-informed hard constraints with mixture-of-experts	[99, 118, 159]
[166]	0.29	Numerical Study of Neural Network Surrogate Model Performance for Uncertainty Propagation	[99, 118]
[168]	0.29	Physics-informed neural operator for forecasting vehicle-induced bridge vibration trajectories	[99, 118]
[169]	0.29	A multi-scale physics-informed convolutional framework with adaptive loss design for parametric steady-state PDEs	[99, 118]
[170]	0.29	Extended parametric physics-informed neural networks (ePPINNs) for efficient solution of heat transfer and fluid flow problems	[99, 118]
[171]	0.29	Neural Quantum Spectral Operator Learning for Solving Partial Differential Equations	[81, 118]
[3]	0.28	Calibrated Physics-Informed Uncertainty Quantification	[35, 42, 81, 99, 106, 109, 116, 118, 123]
[14]	0.28	Learn and Verify: A Framework for Rigorous Verification of Physics-Informed Neural Networks	[99, 101, 115, 159]
[54]	0.27	Data-Efficient Neural Operator Training via Physics-Based Active Learning	[99, 115, 151, 159]
[172]	0.24	Physics-informed Machine Learning for Prognostics and Health Management: Foundations, Advances and Prospects	[35, 118]

[48]	0.23	Hybrid Neural World Models	[99, 115, 159]
[16]	0.22	Diagnosing Failure Modes of Neural Operators Across Diverse PDE Families	[101, 115, 118, 159]
[118]	0.21	Physics-Informed Neural Operator for Learning Partial Differential Equations	[99, 159]

References

- [1] [Research on Multi-scenario Collaborative Testing Methods for Intelligent Algorithms in Physical Field Prediction](#)
Lianzhi Qi, ..., and Yafen Shen. Proceedings of the 2025 5th International Conference on Computational Modeling, Simulation and Data Analysis, 2025. 0 citations.
100% Topic Match
Proposes a hierarchical, scenario-driven metamorphic testing framework for physical-field ML models.
Builds four scenario types (benchmark/extreme/perturbation/cross domain) and constraint rule coupled metamorphic relations based on multi parameter physical correlations and statistics.
Relevant: emphasizes physics-derived OOD transforms, metamorphic oracles, and multi-attribute violation-based assessment, but paper appears methodological/demo on cylinder flow; unclear if it quantifies relation violation rates, encodes relation domain validity, or maps reliable operating boundaries.
- [2] [Research on Mutation Testing Strategies for Intelligent Models Predicting Fluid Velocity Fields](#)
Ke Yu, ..., and Lianzhi Qi. 2025 7th International Academic Exchange Conference on Science and Technology Innovation (IAECST), 2025. 0 citations.
99% Topic Match
Proposes a metamorphic testing method integrating physics-based invariants.
Constructs disturbance and cross-domain transforms using Reynolds-number conservation and Strouhal-number correlations as test relations.
Claims quantitative evaluation of physical consistency and extrapolation; unclear if relation domains, violation-rate statistics, discretization/tolerance handling, or mapping to reliable operating boundaries are explicitly addressed.
- [3] [Calibrated Physics-Informed Uncertainty Quantification](#)
Vignesh Gopakumar, ..., and M. Deisenroth. ArXiv, 2025. 7 citations.
87% Topic Match
Proposes a physics-informed conformal prediction framework for calibrated, data-free UQ of neural PDEs.
Uses convolutional layers as finite-difference stencils and physics residuals as nonconformity scores to get marginal/joint coverage guarantees without labels.
Relevant: focuses on executable relation (PDE residual) as oracle and calibration of violation/statistical coverage, but does not emphasize metamorphic OOD transformations, relation-domain validity handling, or mapping violation rates to operating boundaries.
- [4] [Testing CPS With Design Assumptions-Based Metamorphic Relations and Genetic Programming](#)
C. Mandrioli, ..., and Lionel C. Briand. IEEE Transactions on Software Engineering, 2024. 2 citations.
76% Topic Match
Proposes design-assumption-based metamorphic testing for CPS using genetic programming.
Defines metamorphic relations from control-theoretic design assumptions and evolves input traces + expected outputs via genetic programming.
Relevant as a physics-grounded relation/oracle approach and OOD test generation, but focuses on CPS control traces (not PDE/PINN/neural-operator models) and lacks explicit handling of PDE residuals, discretization, or domain-of-validity for physical relations.
- [5] [Towards Automated Metamorphic Test Identification for Ocean System Models](#)
D. J. Hiremath, ..., and W. Rath. 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing (MET), 2021. 7 citations.
42% Topic Match
Proposes an automated ML-based method to identify metamorphic test relations for ocean system models.
Uses a cost-function search over identity-map representations, ML optimization, dimensionality reduction, and targeted attribute mutation to generate relations.
Focuses on software-level metamorphic relation discovery for geophysical models; does not connect to physics-based V&V thresholds, relation-domain validity, or violation-rate-based operating envelopes.
- [6] [Increasing Validity of Simulation Models Through Metamorphic Testing](#)
Megan M. Olsen and M. Raunak. IEEE Transactions on Reliability, 2019. 27 citations.
37% Topic Match
Proposes applying metamorphic testing to validate executable simulation models.
Adapts MT pseudo-oracles (metamorphic relations) and gives a framework plus guidelines for ABM and DES, demonstrated with three case studies.
Relevant as a general metamorphic-testing 'simulation-V&V bridge, but it does not target scientific-ML, PDEs, PINNs, neural operators, physics-based residuals, OOD physical transforms, or relation-domain validity tied to governing equations/discretization.
- [7] [XRePIT: A deep learning-computational fluid dynamics hybrid framework implemented in OpenFOAM for fast, robust, and scalable unsteady simulations](#)
Shilaj Baral, ..., and Joongoo Jeon. ArXiv, 2025. 1 citations.
34% Topic Match
Proposes a hybrid ML-CFD framework (XRePIT) that switches between neural surrogate and OpenFOAM solver based on residual thresholds.
Implements an automated residual-guided coupling in OpenFOAM, demonstrating stable long-term 3D buoyancy-driven flow simulations and ~2.9x speedup with L2 errors H1e-3.
Relevant because it uses executable PDE residuals as metamorphic-style monitors to trigger solver fallback; but it does not appear to quantify relation-level violation rates across parameter space or explicitly handle domain-of-validity annotations for each relation.
- [8] [Time-marching neural operator-FE coupling: AI-accelerated physics modeling](#)
Wei Wang, ..., and Somdatta Goswami. Computer Methods in Applied Mechanics and Engineering, 2025. 12 citations.
22% Topic Match
Proposes a hybrid time-marching DeepONet-FEM coupling for accelerated PDE simulation.
Achieves this via Schwarz domain-decomposition coupling, embedding a time-stepping scheme inside DeepONet, and adaptive ML-resolved subdomain evolution.
Relevant: targets learned operators and learned simulators, addresses long-time error accumulation and adaptive OOD regions, but paper appears focused on coupling/efficiency and reports error/convergence metrics rather than systematic metamorphic/physics-based oracles, relation-level violation rates, or explicit domain-of-validity annotations.
- [9] [Metamorphic Testing: A Simple Yet Effective Approach for Testing Scientific Software](#)
Upulee Kanewala and T. Chen. Computing in Science & Engineering, 2019. 15 citations.
19% Topic Match
Proposes metamorphic testing (MT) as an effective approach for testing scientific software.
Explains MT principles, demonstrates how to construct metamorphic relations and apply MT without needing explicit oracles.
Relevant as a foundational, general-methods reference on relation-based oracles, but lacks ML/PINN-specific, physics-derived OOD transforms, violation-rate operating envelopes, or domain-of-validity treatment.
- [10] [Physics-Informed AI in Python: Neural Operators with Verified Error Bounds](#)
Murali Krishna Pasupuleti. International Journal of Academic and Industrial Research Innovations(IJAIRI), 2023. 0 citations.
15% Topic Match
Proposes a framework for physics-informed neural operators with instance-wise certified error bounds.
Combines Fourier/DeepONet backbones trained with PDE residuals, energy and BC penalties, plus Lipschitz estimation, interval bound propagation, and conformal calibration to produce certified prediction sets.

Relevant for verified-surrogate V&V (operator learners + executable bounds), but focuses on certification and UQ rather than metamorphic/property-based or physically derived OOD transformation testing or relation-violation-based operating envelopes; notes conservatism in strongly nonlinear/shock regimes and computational cost for tight Lipschitz bounds.

[11] [Quantum-Inspired PINNs in MATLAB: Certified Neural PDE Solvers with Error Bound](#)

Murali Krishna Pasupuleti. International Journal of Academic and Industrial Research Innovations(IJAIRI), 2023. 0 citations.

12% Topic Match

Proposes a MATLAB PINN framework with certified L2 error bounds.

Uses a quantum inspired optimizer (Ising/QAOA restart schedules) plus interval bound propagation and posterior residual analysis to produce tight certified bounds on synthetic PDEs.

Relevant aspects: focuses on post-training certification (separate from training loss), links collocation density/network capacity/optimizer to bounds, reports V&V-style metrics — but does not discuss metamorphic/property based oracles, physics derived OOD transforms, relation violation rates, or explicit domain of validity handling.

[12] [Metamorphic Testing for Hybrid Simulation Validation](#)

Mohammed Farhan, ..., and M. Raunak. 2021 Annual Modeling and Simulation Conference (ANNSIM), 2021. 3 citations.

11% Topic Match

Abstract: Proper validation of a simulation model is essential for confidence in its accuracy and credibility. However, many of the most effective approaches for simulation validation require access to data that may be unavailable. Metamorphic Testing (MT), an approach from traditional software testing, has been shown to be useful for verification of simulation software in similar situations. Recent research shows that MT can be applied to the validation of agent based and discrete event simulation models as well. In this paper we build on that work and show how MT can be applied to hybrid simulation models. We demonstrate the effectiveness...

[13] [Contextual Understanding and Improvement of Metamorphic Testing in Scientific Software Development](#)

Zedong Peng, ..., and Nan Niu. Proceedings of the 15th ACM / IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM), 2021. 10 citations.

10% Topic Match

Abstract: Background: Metamorphic testing emerges as a simple and effective approach for testing scientific software; yet, its adoption in actual scientific software projects is less studied. Aims: In order for the practitioners to better adopt metamorphic testing in their projects, we set out to first gain a deep understanding about the current quality assurance workflow, testing practices, and tools. Method: We propose to integrate various empirical sources, including artifact analysis, stakeholder interviews, and gap analysis from the literature. Results: Applying our approach to the Open Water Analytics Stormwater Management Model project helped to identify four new needs requiring continued and more...

[14] [Learn and Verify: A Framework for Rigorous Verification of Physics-Informed Neural Networks](#)

Kazuaki Tanaka and Kohei Yatabe. ArXiv, 2026. 1 citations.

9% Topic Match

Abstract: The numerical solution of differential equations using neural networks has become a central topic in scientific computing, with Physics-Informed Neural Networks (PINNs) emerging as a powerful paradigm for both forward and inverse problems. However, unlike classical numerical methods that offer established convergence guarantees, neural network-based approximations typically lack rigorous error bounds. Furthermore, the non-deterministic nature of their optimization makes it difficult to mathematically certify their accuracy. To address these challenges, we propose a "Learn and Verify" framework that provides computable, mathematically rigorous error bounds for the solutions of differential equations. By combining a novel Doubly Smoothed Maximum (DSM) loss for training with...

[15] [Identifying metamorphic relations: A data mutation directed approach](#)

Chang-ai Sun, ..., and W. Chan. Software: Practice and Experience, 2023. 4 citations.

8% Topic Match

Abstract: Metamorphic testing (MT) is an effective technique to alleviate the test oracle problem. The principle of MT is to detect failures by checking whether some necessary properties, commonly known as metamorphic relations (MRs), of software under test (SUT) hold among multiple executions of source and follow up test cases. Since both the generation of follow up test cases and test result verification depend on MRs, the identification of MRs plays a key role in MT, which is an important yet difficult task requiring deep domain knowledge of the SUT. Accordingly, techniques that can direct a tester to identify MRs effectively are desirable....

[16] [Diagnosing Failure Modes of Neural Operators Across Diverse PDE Families](#)

Lennon Shikhman. ArXiv, 2026. 2 citations.

7% Topic Match

Abstract: Neural PDE solvers are increasingly used as learned surrogates for families of partial differential equations, where the key machine learning challenge is not only interpolation on a fixed benchmark distribution but generalization under structured shifts in coefficients, boundary conditions, discretization, and rollout horizon. Yet evaluation is still often dominated by in-distribution test error, making robustness difficult to assess. We introduce a standardized stress-testing framework for neural PDE solvers under deployment-relevant shift. We instantiate it on three representative architectures -- Fourier Neural Operators (FNOs), a DeepONet-style model, and convolutional neural operators (CNOs) -- across five qualitatively different PDE families: dispersive, elliptic,...

[17] [Exploratory Metamorphic Testing for Scientific Software](#)

Xuanyi Lin, ..., and D. Rouson. Computing in Science & Engineering, 2018. 29 citations.

6% Topic Match

Abstract: Scientific model developers are able to verify and validate their software via metamorphic testing (MT), even when the expected output of a given test case is not readily available. The tenet is to check whether certain relations hold among the expected outputs of multiple-related inputs. Contemporary approaches require the relations to be defined before tests. Our experience shows that it is often straightforward to first define the multiple iterations of tests for performing continuous simulations, and then keep multiple and even competing metamorphic relations open for investigating the testing-result patterns. We call this new approach exploratory MT, and report our...

[18] [Uncertainty quantification of surrogate models using conformal prediction](#)

Vignesh Gopakumar, ..., and M. Deisenroth. Machine Learning: Science and Technology, 2024. 11 citations.

6% Topic Match

Abstract: Data-driven surrogate models offer fast, inexpensive approximations to complex numerical and experimental systems but typically lack uncertainty quantification, limiting their reliability in safety-critical applications. While Bayesian methods provide uncertainty estimates, they offer no statistical guarantees and struggle with high-dimensional spatio-temporal problems due to computational costs and dependence on prior specification. We present a conformal prediction (CP) framework that provides statistically guaranteed marginal coverage for surrogate models in a model-agnostic manner with near-zero computational cost. Our approach handles high-dimensional spatio-temporal outputs by performing cell-wise calibration while preserving the tensorial structure of predictions. Through extensive empirical evaluation across diverse applications-including partial differential...

[19] [Exact and approximate error bounds for physics-informed neural networks](#)

Augusto T. Chantada, ..., and C. Sc'occola. ArXiv, 2024. 3 citations.

5% Topic Match

Abstract: The use of neural networks to solve differential equations, as an alternative to traditional numerical solvers, has increased recently. However, error bounds for the obtained solutions have only been developed for certain equations. In this work, we report important progress in calculating error bounds of physics-informed neural networks (PINNs) solutions of nonlinear first-order ODEs. We give a general expression that describes the error of the solution that the PINN-based method provides for a nonlinear first-order ODE. In addition, we propose a technique to calculate an approximate bound for the general case and an exact bound for a particular case. The...

[20] [Physics-consistent machine learning with output projection onto physical manifolds](#)

M. Valente, ..., and Rodrigo Ventura. Communications Physics, 2025. 11 citations.

5% Topic Match

Abstract: Data-driven machine learning models often require extensive datasets, which can be costly or inaccessible, and their predictions may fail to comply with established physical laws. Current approaches for incorporating physical priors mitigate these issues by penalizing deviations from known physical laws, as in physics-informed neural networks, or by designing architectures that automatically satisfy specific invariants. However, penalization approaches do not guarantee

compliance with physical constraints for unseen inputs, and invariant-based methods lack flexibility and generality. We propose a physics-consistent machine learning method that directly enforces compliance with physical principles by projecting model outputs onto the manifold defined by these laws....

[21] [Metamorphic Testing on the Continuum of Verification and Validation of Simulation Models](#)

M. Raunak and Megan M. Olsen. 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing (MET), 2021. 8 citations.

4% Topic Match

Abstract: Metamorphic testing has been shown to be useful in testing "non-testable" programs in many domains. Modeling & simulation is one such domain, where both verification and validation can be difficult due to lack of oracles. Although the definition of verification and validation vary slightly in modeling and simulation when compared to standard software, we show that metamorphic testing is appropriate in both aspects of ensuring that a simulation model is accurate. In this paper we expand on our five years of prior work on metamorphic testing for simulation validation to show how metamorphic testing can be used for verification as...

[22] [Metamorphic Testing of AI-based Applications: A Critical Review](#)

Muhammad Nadeem Khokhar, ..., and Muhammad Fiaz. International Journal of Advanced Computer Science and Applications, 2020. 4 citations.

4% Topic Match

Abstract: Metamorphic testing is the youngest testing approach among other members of the testing family. It is designed to test software, which are complex in nature and it is difficult to compute test oracle for them against a given set of inputs. Metamorphic testing approach tests the software with the help of metamorphic relations that guide the tester to check if the observed output can be produced after applying a certain input. Since its first appearance, a lot of research has been done to check its effectiveness on different complex families of software applications like search engines, compilers, artificial intelligence (AI)...

[23] [Verifiable Error Bounds for Physics-Informed Neural Network Solutions of Lyapunov and Hamilton–Jacobi–Bellman Equations](#)

Jun Liu. IEEE Control Systems Letters, 2026. 0 citations.

3% Topic Match

Abstract: Many core problems in nonlinear systems analysis and control can be recast as solving partial differential equations (PDEs) such as Lyapunov and Hamilton–Jacobi–Bellman (HJB) equations. Physics-informed neural networks (PINNs) have emerged as a promising mesh-free approach for approximating their solutions, but in most existing works there is no rigorous guarantee that a small PDE residual implies a small solution error. This letter develops verifiable error bounds for approximate solutions of Lyapunov and HJB equations, with particular emphasis on PINN-based approximations. For both the Lyapunov and HJB PDEs, we show that a verifiable residual bound yields relative error bounds with respect...

[24] [Physics-Constrained Generative Adversarial Networks for 3D Turbulence](#)

D. Tretiak, ..., and D. Livescu. ArXiv, 2022. 4 citations.

3% Topic Match

Abstract: Generative Adversarial Networks (GANs) have received wide acclaim among the machine learning (ML) community for their ability to generate realistic 2D images. ML is being applied more often to complex problems beyond those of computer vision. However, current frameworks often serve as black boxes and lack physics embeddings, leading to poor ability in enforcing constraints and unreliable models. In this work, we develop physics embeddings that can be stringently imposed, referred to as hard constraints, in the neural network architecture. We demonstrate their capability for 3D turbulence by embedding them in GANs, particularly to enforce the mass conservation constraint in...

[25] [Discovering Metamorphic Relations for Scientific Software From User Forums](#)

Xuanyi Lin, ..., and Karla Morris. Computing in Science & Engineering, 2021. 5 citations.

3% Topic Match

Abstract: Scientific software can be used for decades and is constantly evolving. Recently, metamorphic testing, a property-based testing technique, has shown to be effective in testing scientific software, and the necessary properties are expressed as metamorphic relations. However, the development of metamorphic relations is difficult: it requires considerable practical expertise for the software tester. In this article, we report our experience of uncovering metamorphic relations from a user forum's questions of the United States Environmental Protection Agency's Storm Water Management Model (SWMM). Our study not only illustrates a wealth of end users' expertise in interpreting software results, but also demonstrates the...

[26] [Testing Scientific Software with Invariant Relations: A Case Study](#)

Junhua Ding, ..., and Xin-Hua Hu. 2019 IEEE 19th International Conference on Software Quality, Reliability and Security (QRS), 2019. 4 citations.

2% Topic Match

Abstract: Adequately testing scientific software is essential to the quality of the software. However, it is a grand challenge due to the oracle problem. Metamorphic testing has shown its effectiveness for alleviating the problem. But the effectiveness of metamorphic testing is highly dependent on the quality of metamorphic relations that are developed for testing the software. In this paper, we propose a framework for iteratively developing metamorphic relations for adequately testing scientific software. The basic idea is to refine metamorphic relations that are loosely defined to those that can be verified with only limited number of cases so that the relations...

[27] [Rigorous Error Certification for Neural PDE Solvers: From Empirical Residuals to Solution Guarantees](#)

Amartya Mukherjee, ..., and Jun Liu. ArXiv, 2026. 0 citations.

2% Topic Match

Abstract: Uncertainty quantification for partial differential equations is traditionally grounded in discretization theory, where solution error is controlled via mesh/grid refinement. Physics-informed neural networks fundamentally depart from this paradigm: they approximate solutions by minimizing residual losses at collocation points, introducing new sources of error arising from optimization, sampling, representation, and overfitting. As a result, the generalization error in the solution space remains an open problem. Our main theoretical contribution establishes generalization bounds that connect residual control to solution-space error. We prove that when neural approximations lie in a compact subset of the solution space, vanishing residual error guarantees convergence to the...

[28] [Benchmarking neural surrogates on realistic spatiotemporal multiphysics flows](#)

Runze Mao, ..., and Zhi X. Chen. ArXiv, 2025. 1 citations.

2% Topic Match

Abstract: Predicting multiphysics dynamics is computationally expensive and challenging due to the severe coupling of multi-scale, heterogeneous physical processes. While neural surrogates promise a paradigm shift, the field currently suffers from an "illusion of mastery", as repeatedly emphasized in top-tier commentaries: existing evaluations overly rely on simplified, low-dimensional proxies, which fail to expose the models' inherent fragility in realistic regimes. To bridge this critical gap, we present REALM (REalistic AI Learning for Multiphysics), a rigorous benchmarking framework designed to test neural surrogates on challenging, application-driven reactive flows. REALM features 11 high-fidelity datasets spanning from canonical multiphysics problems to complex propulsion and fire safety...

[29] [Benchmarking autoregressive conditional diffusion models for turbulent flow simulation](#)

Georg Kohl, ..., and Nils Thuerey. Neural networks : the official journal of the International Neural Network Society, 2023. 68 citations.

2% Topic Match

Abstract: Simulating turbulent flows is crucial for a wide range of applications, and machine learning-based solvers are gaining increasing relevance. However, achieving temporal stability when generalizing to longer rollout horizons remains a persistent challenge for learned PDE solvers. In this work, we analyze if fully data-driven fluid solvers that utilize an autoregressive rollout based on conditional diffusion models are a viable option to address this challenge. We investigate accuracy, posterior sampling, spectral behavior, and temporal stability, while requiring that methods generalize to flow parameters beyond the training regime. To quantitatively and qualitatively benchmark the performance of various flow prediction approaches, three...

[30] [Frequency Bias and OOD Generalization in Neural Operators under a Variable-Coefficient Wave Equation](#)

Runlong Xie and Andrew F. Luo. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Neural operators learn to map initial conditions to the terminal solution of partial differential equations (PDEs), providing a surrogate for the full operator mapping. This enables rapid prediction across different input configurations. While recent neural operator architectures have demonstrated strong performance on diverse PDE tasks, their behavior under structured distribution shifts remains insufficiently understood. To investigate this, we study operator learning in a wave propagation setting governed by a one-dimensional variable-coefficient wave equation, using two representative architectures, the Fourier Neural Operator (FNO) and the Deep Operator Network (DeepONet). To examine their generalization under distribution shifts, we consider structured out-of-distribution (OOD)...

[31] [Residual-based error bound for physics-informed neural networks](#)

Shuheng Liu, ..., and P. Protopapas. ArXiv, 2023. 11 citations.

1% Topic Match

Abstract: Neural networks are universal approximators and are studied for their use in solving differential equations. However, a major criticism is the lack of error bounds for obtained solutions. This paper proposes a technique to rigorously evaluate the error bound of Physics-Informed Neural Networks (PINNs) on most linear ordinary differential equations (ODEs), certain nonlinear ODEs, and first-order linear partial differential equations (PDEs). The error bound is based purely on equation structure and residual information and does not depend on assumptions of how well the networks are trained. We propose algorithms that bound the error efficiently. Some proposed algorithms provide tighter bounds...

[32] [Application of Property-based Testing Tools for Metamorphic Testing](#)

Nasser Alzahrani, ..., and James Harland. ArXiv, 2022. 2 citations.

1% Topic Match

Abstract: Metamorphic testing (MT) is a general approach for the testing of a specific kind of software systems -- so-called "non-testable", where the "classical" testing approaches are difficult to apply. MT is an effective approach for addressing the test oracle problem and test case generation problem. The test oracle problem is when it is difficult to determine the correct expected output of a particular test case or to determine whether the actual outputs agree with the expected outcomes. The core concept in MT is metamorphic relations (MRs) which provide formal specification of the system under test. One of the challenges in...

[33] [A numerical study into neural network surrogate model performance for uncertainty propagation](#)

Noah Wade and K. Teferra. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Neural network surrogate models have emerged as a promising approach to model solution fields for a wide variety of boundary value problems encountered in physical modeling. Stochastic problems represent an area of particularly high interest because of the potential to significantly reduce the repeated evaluation of expensive forward models via traditional numerical solvers when conducting parametric analysis. However, many studies found in the literature primarily focus on the ability of neural network surrogate models to represent deterministic samples or mean field solutions and largely overlook surrogate model performance at the tails of the distribution. The present study examines in detail...

[34] [An RBC-MsUQ Framework for Red Blood Cell Morpho-Mechanics](#)

Shuo Wang, ..., and Tao Zhou. ArXiv, 2025. 0 citations.

1% Topic Match

Abstract: Characterizing the morpho-mechanical properties of red blood cells (RBCs) is crucial for understanding microvascular transport mechanisms and cellular pathophysiological processes, yet current computational models are constrained by multi-source uncertainties including cross-platform experimental discrepancies and parameter identification stochasticity. We present RBC-MsUQ, a novel multi-stage uncertainty quantification framework tailored for RBCs. It integrates hierarchical Bayesian inference with diverse experimental datasets, establishing prior distributions for RBC parameters via microscopic simulations and literature-derived data. A dynamic annealing technique defines stress-free baselines, while deep neural network surrogates, optimized through sensitivity analysis, achieve sub-10% prediction errors for efficient simulation approximation. Its two-stage hierarchical inference architecture constrains...

[35] [Scaling physics-informed hard constraints with mixture-of-experts](#)

N. Chalapathi, ..., and Aditi Krishnapriyan. ArXiv, 2024. 32 citations.

1% Topic Match

Abstract: Imposing known physical constraints, such as conservation laws, during neural network training introduces an inductive bias that can improve accuracy, reliability, convergence, and data efficiency for modeling physical dynamics. While such constraints can be softly imposed via loss function penalties, recent advancements in differentiable physics and optimization improve performance by incorporating PDE-constrained optimization as individual layers in neural networks. This enables a stricter adherence to physical constraints. However, imposing hard constraints significantly increases computational and memory costs, especially for complex dynamical systems. This is because it requires solving an optimization problem over a large number of points in a mesh,...

[36] [Construction method for general phenomenological RANS turbulence model](#)

Shuming Zhang, ..., and Z. Tao. ArXiv, 2021. 0 citations.

1% Topic Match

Abstract: This paper proposes a phenomenological Reynolds Averaged Navier-Stokes (RANS) calculation model based on physical constraints. In this model part of the source terms in the μ equation was replaced with the deep learning model, using the standard $kk-\mu$ model as a template. The simulation results of this model achieved a high error reduction of 51.7 % compared to the standard $kk-\mu$ model. To improve the adaptability and accuracy compared to the convergence of the abnormal flow regime, the coordinate technology proposed in this study was used in the modelling process. For the training data, the $kk-\mu$...

[37] [Metamorphic Testing in Autonomous System Simulations](#)

Jubril Gbolahan Adigun, ..., and M. Felderer. 2022 48th Euromicro Conference on Software Engineering and Advanced Applications (SEAA), 2022. 3 citations.

1% Topic Match

Abstract: Metamorphic testing has proven to be effective for test case generation and fault detection in many domains. It is a software testing strategy that uses certain relations between input-output pairs of a program, referred to as metamorphic relations. This approach is relevant in the autonomous systems domain since it helps in cases where the outcome of a given test input may be difficult to determine. In this paper therefore, we provide an overview of metamorphic testing as well as an implementation in the autonomous systems domain. We implement an obstacle detection and avoidance task in autonomous drones utilising the GNC...

[38] [Evaluating Error Bound for Physics-Informed Neural Networks on Linear Dynamical Systems](#)

Shuheng Liu, ..., and P. Protopapas. ArXiv, 2022. 6 citations.

1% Topic Match

Abstract: There have been extensive studies on solving differential equations using physics-informed neural networks. While this method has proven advantageous in many cases, a major criticism lies in its lack of analytical error bounds. Therefore, it is less credible than its traditional counterparts, such as the finite difference method. This paper shows that one can mathematically derive explicit error bounds for physics-informed neural networks trained on a class of linear systems of differential equations. More importantly, evaluating such error bounds only requires evaluating the differential equation residual infinity norm over the domain of interest. Our work shows a link between network...

[39] [Quantification of total uncertainty in the physics-informed reconstruction of CVSim-6 physiology](#)

Mario De Florio, ..., and G. Karniadakis. ArXiv, 2024. 14 citations.

0% Topic Match

Abstract: When predicting physical phenomena through simulation, quantification of the total uncertainty due to multiple sources is as crucial as making sure the underlying numerical model is accurate. Possible sources include irreducible aleatoric uncertainty due to noise in the data, epistemic uncertainty induced by insufficient data or inadequate parameterization, and model-form uncertainty related to the use of misspecified model equations. In addition, recently proposed approaches provide flexible ways to combine information from data with full or partial satisfaction of equations that typically encode physical principles. Physics-based regularization interacts in nontrivial ways with aleatoric, epistemic and model-form uncertainty and their combination, and...

[40] [A Survey on Metamorphic Testing](#)

Sergio Segura, ..., and Antonio Ruiz-Cortés. IEEE Transactions on Software Engineering, 2016. 529 citations.

0% Topic Match

Abstract: A test oracle determines whether a test execution reveals a fault, often by comparing the observed program output to the expected output. This is not always practical, for example when a program's input-output relation is complex and difficult to capture formally. Metamorphic testing provides an alternative, where correctness is not determined by checking an individual concrete output, but by applying a transformation to a test input and observing how the program output "morphs" into a different one as a result. Since the introduction of such metamorphic relations in 1998, many contributions on metamorphic testing have been made, and the technique...

[41] [Discovering a reaction-diffusion model for Alzheimer's disease by combining PINNs with symbolic regression](#)

Zhen Zhang, ..., and G. Karniadakis. ArXiv, 2023. 68 citations.

0% Topic Match

Abstract: Misfolded tau proteins play a critical role in the progression and pathology of Alzheimer's disease. Recent studies suggest that the spatio-temporal pattern of misfolded tau follows a reaction-diffusion type equation. However, the precise mathematical model and parameters that characterize the progression of misfolded protein across the brain remain incompletely understood. Here, we use deep learning and artificial intelligence to discover a mathematical model for the progression of Alzheimer's disease using longitudinal tau positron emission tomography from the Alzheimer's Disease Neuroimaging Initiative database. Specifically, we integrate physics informed neural networks (PINNs) and symbolic regression to discover a reaction-diffusion type partial differential...

[42] [Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons](#)

Apostolos F. Psaros, ..., and G. Karniadakis. J. Comput. Phys., 2022. 423 citations.

0% Topic Match

Abstract: Neural networks (NNs) are currently changing the computational paradigm on how to combine data with mathematical laws in physics and engineering in a profound way, tackling challenging inverse and ill-posed problems not solvable with traditional methods. However, quantifying errors and uncertainties in NN-based inference is more complicated than in traditional methods. This is because in addition to aleatoric uncertainty associated with noisy data, there is also uncertainty due to limited data, but also due to NN hyperparameters, overparametrization, optimization and sampling errors as well as model misspecification. Although there are some recent works on uncertainty quantification (UQ) in NNs, there...

[43] [Diffusion model based data generation for partial differential equations](#)

Rucha Apte, ..., and Jay Pathak. ArXiv, 2023. 6 citations.

0% Topic Match

Abstract: In a preliminary attempt to address the problem of data scarcity in physics-based machine learning, we introduce a novel methodology for data generation in physics-based simulations. Our motivation is to overcome the limitations posed by the limited availability of numerical data. To achieve this, we leverage a diffusion model that allows us to generate synthetic data samples and test them for two canonical cases: (a) the steady 2-D Poisson equation, and (b) the forced unsteady 2-D Navier-Stokes (NS) {vorticity-transport} equation in a confined box. By comparing the generated data samples against outputs from classical solvers, we assess their accuracy and...

[44] [Simulator-free solution of high-dimensional stochastic elliptic partial differential equations using deep neural networks](#)

Sharmila Karumuri, ..., and Jitesh H. Panchal. J. Comput. Phys., 2019. 154 citations.

0% Topic Match

Abstract: Abstract Stochastic partial differential equations (SPDEs) are ubiquitous in engineering and computational sciences. The stochasticity arises as a consequence of uncertainty in input parameters, constitutive relations, initial/boundary conditions, etc. Because of these functional uncertainties, the stochastic parameter space is often high-dimensional, requiring hundreds, or even thousands, of parameters to describe it. This poses an insurmountable challenge to response surface modeling since the number of forward model evaluations needed to construct an accurate surrogate grows exponentially with the dimension of the uncertain parameter space; a phenomenon referred to as the curse of dimensionality. State-of-the-art methods for high-dimensional uncertainty propagation seek to...

[45] [Bayesian Hidden Physics Models: Uncertainty Quantification for Discovery of Nonlinear Partial Differential Operators from Data](#)

Steven Atkinson. ArXiv, 2020. 10 citations.

0% Topic Match

Abstract: What do data tell us about physics-and what don't they tell us? There has been a surge of interest in using machine learning models to discover governing physical laws such as differential equations from data, but current methods lack uncertainty quantification to communicate their credibility. This work addresses this shortcoming from a Bayesian perspective. We introduce a novel model comprising "leaf" modules that learn to represent distinct experiments' spatiotemporal functional data as neural networks and a single "root" module that expresses a nonparametric distribution over their governing nonlinear differential operator as a Gaussian process. Automatic differentiation is used to compute...

[46] [Representation Equivalent Neural Operators: a Framework for Alias-free Operator Learning](#)

Francesca Bartolucci, ..., and Rima Alaifari. Advances in Neural Information Processing Systems 36, 2023. 66 citations.

0% Topic Match

Abstract: Recently, operator learning, or learning mappings between infinite-dimensional function spaces, has garnered significant attention, notably in relation to learning partial differential equations from data. Conceptually clear when outlined on paper, neural operators necessitate discretization in the transition to computer implementations. This step can compromise their integrity, often causing them to deviate from the underlying operators. This research offers a fresh take on neural operators with a framework Representation equivalent Neural Operators (ReNO) designed to address these issues. At its core is the concept of operator aliasing, which measures inconsistency between neural operators and their discrete representations. We explore this for...

[47] [BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations](#)

Jonathan Gorard, ..., and J. Juno. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: The traditional limitations of neural networks in reliably generalizing beyond the convex hulls of their training data present a significant problem for computational physics, in which one often wishes to solve PDEs in regimes far beyond anything which can be experimentally or analytically validated. In this paper, we show how it is possible to circumvent these limitations by constructing formally-verified neural network solvers for PDEs, with rigorous convergence, stability, and conservation properties, whose correctness can therefore be guaranteed even in extrapolatory regimes. By using the method of characteristics to predict the analytical properties of PDE solutions a priori (even in...

[48] [Hybrid Neural World Models](#)

Pranav Lakshmanan and Paras Chopra. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Neural surrogates promise large speedups over classical solvers for physical dynamics but fail silently at sharp dynamical events such as shocks, fronts, and contact. We present hybrid neural world models for physical dynamics: a recipe for training and deploying multi-horizon surrogates in physical state space, where a single network with continuous horizon conditioning is trained with direct supervision against textbook reference solvers to predict any future state at horizon T in one forward pass. Although no part of the training data, loss function, or architecture supervises discontinuity location, the trained surrogate encodes it implicitly, recoverable from its forward passes alone...

[49] [Distribution-Free Robust Predict-Then-Optimize in Function Spaces](#)

Yash Patel and Ambuj Tewari. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: The need to rapidly solve PDEs in engineering design workflows has spurred the rise of neural surrogate models. In particular, neural operator models provide a discretization-invariant surrogate by retaining the infinite-dimensional, functional form of their arguments. Despite improved throughput, such methods lack guarantees on accuracy, unlike classical numerical PDE solvers. Optimizing engineering designs under these potentially miscalibrated surrogates thus runs the risk of producing designs that perform poorly upon deployment. In a similar vein, there is growing interest in automated decision-making under black-box predictors in the finite-dimensional setting, where a similar risk of suboptimality exists under poorly calibrated models. For...

[50] [Research on the Application of Physical Information Neural Network in Multiscale Fluid Dynamics Simulation](#)

Jinyu Liu, ..., and Junhui Wang. International Journal of Computer Science and Information Technology, 2025. 0 citations.

0% Topic Match

Abstract: In this paper, a multi-scale fluid dynamics simulation framework based on physical information neural network (PINN) is proposed, which achieves efficient simulation of multi-scale flow characteristics by means of a multi-resolution network structure, physical constraints on scale decomposition and adaptive sampling strategy. Experimental results show that the framework outperforms traditional methods in microscale flow, turbulence and multiphase flow problems, with a 37% improvement in the average prediction accuracy and a 75% reduction in the computational resource requirement. In particular, it shows excellent generalisation performance when dealing with complex flows with large scale spans, providing a new method for fluid dynamics...

[51] [A Conformal Prediction Framework for Uncertainty Quantification in Physics-Informed Neural Networks](#)

Yifan Yu, ..., and Yangshuai Wang. ArXiv, 2025. 3 citations.

0% Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) have emerged as a powerful framework for solving PDEs, yet existing uncertainty quantification (UQ) approaches for PINNs generally lack rigorous statistical guarantees. In this work, we bridge this gap by introducing a distribution-free conformal prediction (CP)

framework for UQ in PINNs. This framework calibrates prediction intervals by constructing nonconformity scores on a calibration set, thereby yielding distribution-free uncertainty estimates with rigorous finite-sample coverage guarantees for PINNs. To handle spatial heteroskedasticity, we further introduce local conformal quantile estimation, enabling spatially adaptive uncertainty bands while preserving theoretical guarantee. Through systematic evaluations on typical PDEs (damped harmonic oscillator, Poisson,...

[52] [SURF: A Generalization Benchmark for GNNs Predicting Fluid Dynamics](#)

Stefan Künzli, ..., and R. Wattenhofer. ArXiv, 2023. 0 citations.

0% Topic Match

Abstract: Simulating fluid dynamics is crucial for the design and development process, ranging from simple valves to complex turbomachinery. Accurately solving the underlying physical equations is computationally expensive. Therefore, learning-based solvers that model interactions on meshes have gained interest due to their promising speed-ups. However, it is unknown to what extent these models truly understand the underlying physical principles and can generalize rather than interpolate. Generalization is a key requirement for a general-purpose fluid simulator, which should adapt to different topologies, resolutions, or thermodynamic ranges. We propose SURF, a benchmark designed to test the generalization of learned graph-based fluid simulators. SURF...

[53] [Hierarchical Metamorphic Relations for Testing Scientific Software](#)

Xuanyi Lin, ..., and Nan Niu. 2018 IEEE/ACM 13th International Workshop on Software Engineering for Science (SE4Science), 2018. 22 citations.

0% Topic Match

Abstract: Scientist developers have not yet routinely adopted systematic testing techniques to assure software quality. A key challenge is the oracle problem, a situation in which appropriate mechanisms are unavailable for checking if the code produces the expected output when executed using a set of test cases (TCs). Metamorphic testing alleviates the oracle problem by specifying the relationship that a source TC and its follow-up TC shall meet. Such relationships are called metamorphic relations (MRs) which are necessary properties of the intended program's functionality. Existing approaches handle the MRs in a flat manner. This paper introduces a novel way to facilitate...

[54] [Data-Efficient Neural Operator Training via Physics-Based Active Learning](#)

Alicja Polanska, ..., and Stanislas Pamela. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Solving partial differential equations with neural operators significantly reduces computational costs but remains bottlenecked by high training data requirements. Active learning offers a natural framework to mitigate this by selectively acquiring the most informative samples in an iterative manner. We introduce physics-based acquisition - a novel physics-informed active learning algorithm that leverages the partial differential equation residual to guide data selection. We validate the method by presenting numerical experiments for the 1D Burgers equation and the 2D compressible Navier-Stokes equations. We show that, in our experiments, physics-based acquisition consistently outperforms random acquisition and matches the state of the art...

[55] [Certified machine learning: A posteriori error estimation for physics-informed neural networks](#)

Birgit Hillebrecht and B. Unger. 2022 International Joint Conference on Neural Networks (IJCNN), 2022. 25 citations.

0% Topic Match

Abstract: Physics-informed neural networks (PINNs) are one popular approach to incorporate a priori knowledge about physical systems into the learning framework. PINNs are known to be robust for smaller training sets, derive better generalization problems, and are faster to train. In this paper, we show that using PINNs in comparison with purely data-driven neural networks is not only favorable for training performance but allows us to extract significant information on the quality of the approximated solution. Assuming that the underlying differential equation for the PINN training is an ordinary differential equation, we derive a rigorous upper limit on the PINN prediction...

[56] [Modeling the Dynamics of PDE Systems with Physics-Constrained Deep Auto-Regressive Networks](#)

N. Geneva and N. Zabarar. ArXiv, 2019. 339 citations.

0% Topic Match

Abstract: Abstract In recent years, deep learning has proven to be a viable methodology for surrogate modeling and uncertainty quantification for a vast number of physical systems. However, in their traditional form, such models can require a large amount of training data. This is of particular importance for various engineering and scientific applications where data may be extremely expensive to obtain. To overcome this shortcoming, physics-constrained deep learning provides a promising methodology as it only utilizes the governing equations. In this work, we propose a novel auto-regressive dense encoder-decoder convolutional neural network to solve and model non-linear dynamical systems without training...

[57] [A Multimodal Vision Transformer-based Modeling Framework for Prediction of Fluid Flows in Energy Systems](#)

Kiran K. Yalamanchi, ..., and Pinaki Pal. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Computational fluid dynamics (CFD) simulations of complex fluid flows in energy systems are prohibitively expensive due to strong nonlinearities and multiscale-multiphysics interactions. In this work, we present a transformer-based modeling framework for prediction of fluid flows, and demonstrate it for high-pressure gas injection phenomena relevant to reciprocating engines. The approach employs a hierarchical Vision Transformer (SwinV2-UNet) architecture that processes multimodal flow datasets from multi-fidelity simulations. The model architecture is conditioned on auxiliary tokens explicitly encoding the data modality and time increment. Model performance is assessed on two different tasks: (1) spatiotemporal rollouts, where the model autoregressively predicts the flow state...

[58] [An Advanced Physics-Informed Neural Operator for Comprehensive Design Optimization of Highly-Nonlinear Systems: An Aerospace Composites Processing Case Study](#)

Milad Ramezankhani, ..., and Dagnachew Birru. ArXiv, 2024. 19 citations.

0% Topic Match

Abstract: Deep Operator Networks (DeepONets) and their physics-informed variants have shown significant promise in learning mappings between function spaces of partial differential equations, enhancing the generalization of traditional neural networks. However, for highly nonlinear real-world applications like aerospace composites processing, existing models often fail to capture underlying solutions accurately and are typically limited to single input functions, constraining rapid process design development. This paper introduces an advanced physics-informed DeepONet tailored for such complex systems with multiple input functions. Equipped with architectural enhancements like nonlinear decoders and effective training strategies such as curriculum learning and domain decomposition, the proposed model handles high-dimensional...

[59] [ConvPDE-UQ: Convolutional neural networks with quantified uncertainty for heterogeneous elliptic partial differential equations on varied domains](#)

Nick Winovich, ..., and Guang Lin. J. Comput. Phys., 2019. 108 citations.

0% Topic Match

Abstract: Abstract In this work, we introduce the ConvPDE-UQ framework for constructing light-weight numerical solvers for partial differential equations (PDEs) using convolutional neural networks. A theoretical justification for the neural network approximation to partial differential equation solvers on varied domains is established based on the existence and properties of Green's functions. These solvers are able to effectively reduce the computational demands of traditional numerical methods into a single forward-pass of a convolutional network. The network architecture is also designed to predict pointwise Gaussian posterior distributions, with weights trained to minimize the associated negative log-likelihood of the observed solutions. This setup facilitates...

[60] [Evaluating Uncertainty Quantification approaches for Neural PDEs in scientific applications](#)

Vardhan Dongre and G. S. Hora. ArXiv, 2023. 0 citations.

0% Topic Match

Abstract: The accessibility of spatially distributed data, enabled by affordable sensors, field, and numerical experiments, has facilitated the development of data-driven solutions for scientific problems, including climate change, weather prediction, and urban planning. Neural Partial Differential Equations (Neural PDEs), which combine deep learning (DL) techniques with domain expertise (e.g., governing equations) for parameterization, have proven to be effective in capturing valuable correlations within spatiotemporal datasets. However, sparse and noisy measurements coupled with modeling approximation introduce aleatoric and epistemic uncertainties. Therefore, quantifying uncertainties propagated from model inputs to outputs remains a challenge and an essential goal for establishing the trustworthiness of Neural...

[61] [Learning Under Laws: A Constraint-Projected Neural PDE Solver that Eliminates Hallucinations](#)

Mainak Singha. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Neural networks can approximate solutions to partial differential equations, but they often break the very laws they are meant to model-creating mass from nowhere, drifting shocks, or violating conservation and entropy. We address this by training within the laws of physics rather than beside them. Our framework, called Constraint-Projected Learning (CPL), keeps every update physically admissible by projecting network outputs onto the intersection of constraint sets defined by conservation, Rankine-Hugoniot balance, entropy, and positivity. The projection is differentiable and adds only about 10% computational overhead, making it fully compatible with back-propagation. We further stabilize training with total-variation damping (TVD) to...

[62] [Using Metamorphic Relation Violation Regions to Support a Simulation Framework for the Process of Metamorphic Testing](#)

Zhihao Ying, ..., and Z. Zhou. 2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC), 2022. 6 citations.

0% Topic Match

Abstract: Metamorphic testing (MT) has been growing in popularity, but it can still be quite challenging and time-consuming to assess its performance. Typical approaches to performance assessment can require a series of steps, and depend on a variety of factors, often requiring serendipity. This can be a bottleneck for some aspects of MT research. Central to MT, metamorphic relations (MRs) represent necessary properties of the system under test (SUT). In traditional software testing, simulations are often employed to examine and compare the performance of different testing strategies. However, these simulations are typically designed based on the assumed availability (and applicability) of...

[63] [Morephy-Net: An evolutionary multi-objective optimization for replica-exchange-based physics-informed neural operator learning networks](#)

B.-L. Lu, ..., and Guang Lin. Computer Methods in Applied Mechanics and Engineering, 2025. 3 citations.

0% Topic Match

Abstract: We propose an evolutionary Multi-objective Optimization for Replica-Exchange-based Physics-informed operator-learning Networks (Morephy-Net) to solve parametric partial differential equations (PDEs) in noisy data regimes, for both forward prediction and inverse identification. Existing physics-informed neural networks and operator-learning models (e.g., DeepONets and Fourier neural operators) often face three coupled challenges: (i) balancing data/operator and physics residual losses, (ii) maintaining robustness under noisy or sparse observations, and (iii) providing reliable uncertainty quantification. Morephy-Net addresses these issues by integrating: (i) evolutionary multi-objective optimization that treats data/operator and physics residual terms as separate objectives and searches the Pareto front, thereby avoiding ad hoc loss weighting;...

[64] [Evaluation of machine learning architectures on the quantification of epistemic and aleatoric uncertainties in complex dynamical systems](#)

S. Guth, ..., and T. Sapsis. ArXiv, 2023. 3 citations.

0% Topic Match

Abstract: Machine learning methods for the construction of data-driven reduced order model models are used in an increasing variety of engineering domains, especially as a supplement to expensive computational fluid dynamics for design problems. An important check on the reliability of surrogate models is Uncertainty Quantification (UQ), a self assessed estimate of the model error. Accurate UQ allows for cost savings by reducing both the required size of training data sets and the required safety factors, while poor UQ prevents users from confidently relying on model predictions. We examine several machine learning techniques, including both Gaussian processes and a family UQ-augmented...

[65] [Multi-variance replica exchange SGMCMC for inverse and forward problems via Bayesian PINN](#)

Guang Lin, ..., and Zecheng Zhang. J. Comput. Phys., 2022. 31 citations.

0% Topic Match

No summary or abstract available

[66] [Deep neural operators as accurate surrogates for shape optimization](#)

K. Shukla, ..., and G. Karniadakis. Eng. Appl. Artif. Intell., 2024. 93 citations.

0% Topic Match

No summary or abstract available

[67] [Enforcing Imprecise Constraints on Generative Adversarial Networks for Emulating Physical Systems](#)

Zeng Yang, ..., and Heng Xiao. Communications in Computational Physics, 2019. 20 citations.

0% Topic Match

Abstract: Generative adversarial networks (GANs) are initially proposed to generate images by learning from a large number of samples. Recently, GANs have been used to emulate complex physical systems such as turbulent flows. However, a critical question must be answered before GANs can be considered trusted emulators for physical systems: do GANs-generated samples conform to the various physical constraints? These include both deterministic constraints (e.g., conservation laws) and statistical constraints (e.g., energy spectrum in turbulent flows). The latter has been studied in a companion paper (Wu et al. 2019. Enforcing statistical constraints in generative adversarial networks for modeling chaotic dynamical systems...

[68] [Evaluation of machine learning 1 architectures on the quantification of 2 epistemic and aleatoric uncertainties in 3 complex dynamical systems](#)

S. Guth, ..., and T. Sapsis. Journal Not Provided, 2023. 2 citations.

0% Topic Match

No summary or abstract available

[69] [An Evolutionary Multi-objective Optimization for Replica-Exchange-based Physics-informed Operator Learning Network](#)

B.-L. Lu, ..., and Guang Lin. ArXiv, 2025. 1 citations.

0% Topic Match

No summary or abstract available

[70] [Evaluating Uncertainty Quantification approaches for Neural PDEs in scientific application](#)

No author found. Journal Not Provided, Unknown year. 0 citations.

0% Topic Match

No summary or abstract available

[71] [Automated identification of metamorphic test scenarios for an ocean-modeling application](#)

D. J. Hiremath, ..., and W. Rath. 2020 IEEE International Conference On Artificial Intelligence Testing (AITest), 2020. 6 citations.

0% Topic Match

Abstract: Metamorphic testing seeks to validate software in the absence of test oracles. Our application domain is ocean modeling, where test oracles often do not exist, but where symmetries of the simulated physical systems are known. In this short paper we present work in progress for automated generation of metamorphic test scenarios using machine learning. Metamorphic testing may be expressed as $f(g(X))=h(f(X))$ with f being the application under test, with input data X , and with the metamorphic relation (g, h) . Automatically generated metamorphic relations can be used for constructing regression tests, and for comparing different versions of the same software application. Here, we...

[72] [Neural Inverse Operators for Solving PDE Inverse Problems](#)

R. Molinaro, ..., and Siddhartha Mishra. Unknown journal, 2023. 84 citations.

0% Topic Match

Abstract: A large class of inverse problems for PDEs are only well-defined as mappings from operators to functions. Existing operator learning frameworks map functions to functions and need to be modified to learn inverse maps from data. We propose a novel architecture termed Neural Inverse Operators (NIOs) to solve these PDE inverse problems. Motivated by the underlying mathematical structure, NIO is based on a suitable composition of DeepONets and FNOs to approximate mappings from operators to functions. A variety of experiments are presented to demonstrate that NIOs significantly outperform baselines and solve PDE inverse problems robustly, accurately and are several orders...

[73] [Conformalized Quantum DeepONet Ensembles for Scalable Operator Learning with Distribution-Free Uncertainty](#)

Purav Matlia, ..., and Guang Lin. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Operator learning enables fast surrogate modeling of high-dimensional dynamical systems, but existing approaches face two fundamental limitations: quadratic inference complexity and unreliable uncertainty quantification in safety-critical settings. We propose Conformalized Quantum DeepONet Ensembles, a framework that addresses both challenges simultaneously. By leveraging Quantum Orthogonal Neural Networks (QOrthoNNs), we reduce operator inference complexity from $O(n^2)$ to $O(n)$, enabling scalable evaluation over fine discretizations. To provide rigorous uncertainty quantification, we combine ensemble-based epistemic modeling with adaptive conformal prediction, yielding distribution-free coverage guarantees. A key challenge in ensembling is that naive parallelism scales hardware resources linearly with the number of models. We resolve...

[74] [Calibrated Uncertainty Quantification for Operator Learning via Conformal Prediction](#)

Ziqi Ma, ..., and Anima Anandkumar. Trans. Mach. Learn. Res., 2024. 19 citations.

0% Topic Match

Abstract: Operator learning has been increasingly adopted in scientific and engineering applications, many of which require calibrated uncertainty quantification. Since the output of operator learning is a continuous function, quantifying uncertainty simultaneously at all points in the domain is challenging. Current methods consider calibration at a single point or over one scalar function or make strong assumptions such as Gaussianity. We propose a risk-controlling quantile neural operator, a distribution-free, finite-sample functional calibration conformal prediction method. We provide a theoretical calibration guarantee on the coverage rate, defined as the expected percentage of points on the function domain whose true value lies within...

[75] [Quality measures for the evaluation of 1 machine learning architectures on the 2 quantification of epistemic and aleatoric 3 uncertainties in complex dynamical systems](#)

S. Guth, ..., and T. Sapsis. Journal Not Provided, Unknown year. 0 citations.

0% Topic Match

No summary or abstract available

[76] [Physics-aware generative models for turbulent fluid flows through energy-consistent stochastic interpolants](#)

N. T. Mücke and Benjamin Sanderse. ArXiv, 2025. 9 citations.

0% Topic Match

Abstract: Generative models have demonstrated remarkable success in domains such as text, image, and video synthesis. In this work, we explore the application of generative models to fluid dynamics, specifically for turbulence simulation, where classical numerical solvers are computationally expensive. We propose a novel stochastic generative model based on stochastic interpolants, which enables probabilistic forecasting while incorporating physical constraints such as energy stability and divergence-freeness. Unlike conventional stochastic generative models, which are often agnostic to underlying physical laws, our approach embeds energy consistency by making the parameters of the stochastic interpolant learnable coefficients. We evaluate our method on a benchmark turbulence...

[77] [Bilevel optimization for learning hyperparameters: Application to solving PDEs and inverse problems with Gaussian processes](#)

N. Nelsen, ..., and Zongren Zou. ArXiv, 2025. 6 citations.

0% Topic Match

Abstract: Methods for solving scientific computing and inference problems, such as kernel- and neural network-based approaches for partial differential equations (PDEs), inverse problems, and supervised learning tasks, depend crucially on the choice of hyperparameters. Specifically, the efficacy of such methods, and in particular their accuracy, stability, and generalization properties, strongly depends on the choice of hyperparameters. While bilevel optimization offers a principled framework for hyperparameter tuning, its nested optimization structure can be computationally demanding, especially in PDE-constrained contexts. In this paper, we propose an efficient strategy for hyperparameter optimization within the bilevel framework by employing a Gauss-Newton linearization of the inner...

[78] [Semi-supervised invertible neural operators for Bayesian inverse problems](#)

Sebastian Kaltenbach, ..., and P. Koutsourelakis. Computational Mechanics, 2022. 40 citations.

0% Topic Match

Abstract: Neural Operators offer a powerful, data-driven tool for solving parametric PDEs as they can represent maps between infinite-dimensional function spaces. In this work, we employ physics-informed Neural Operators in the context of high-dimensional, Bayesian inverse problems. Traditional solution strategies necessitate an enormous, and frequently infeasible, number of forward model solves, as well as the computation of parametric derivatives. In order to enable efficient solutions, we extend Deep Operator Networks (DeepONets) by employing a RealNVP architecture which yields an invertible and differentiable map between the parametric input and the branch-net output. This allows us to construct accurate approximations of the full...

[79] [Learning Data-Efficient and Generalizable Neural Operators via Fundamental Physics Knowledge](#)

Siyang Ma, ..., and V. Ganesh. ArXiv, 2026. 3 citations.

0% Topic Match

Abstract: Recent advances in scientific machine learning (SciML) have enabled neural operators (NOs) to serve as powerful surrogates for modeling the dynamic evolution of physical systems governed by partial differential equations (PDEs). While existing approaches focus primarily on learning simulations from the target PDE, they often overlook more fundamental physical principles underlying these equations. Inspired by how numerical solvers are compatible with simulations of different settings of PDEs, we propose a multiphysics training framework that jointly learns from both the original PDEs and their simplified basic forms. Our framework enhances data efficiency, reduces predictive errors, and improves out-of-distribution (OOD) generalization, particularly...

[80] [Realizability-Constrained Machine Learning for Turbulence Closures in Wake Flows](#)

Talib Ansari, ..., and Harshal D. Akolekar. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Computational fluid dynamics (CFD)-driven machine learning frameworks based on symbolic regression offer a promising pathway for turbulence model discovery, but are often hindered by numerical instability, residual stagnation, and non-physical model behavior during training. In particular, realizability, which is rarely enforced explicitly during model development, remains a critical yet overlooked requirement, especially for accurate wake prediction. In this work, a residual- and realizability-filtered CFD-driven framework is proposed to enhance both efficiency and robustness within a gene expression programming (GEP) paradigm. The method integrates two residual-based filtering criteria along with a barycentric-map-based realizability constraint directly into the CFD solution loop, enabling...

[81] [NeuralUQ: A comprehensive library for uncertainty quantification in neural differential equations and operators](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2022. 54 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) in machine learning is currently drawing increasing research interest, driven by the rapid deployment of deep neural networks across different fields, such as computer vision, natural language processing, and the need for reliable tools in risk-sensitive applications. Recently, various machine learning models have also been developed to tackle problems in the field of scientific computing with applications to computational science and engineering (CSE). Physics-informed neural networks and deep operator networks are two such models for solving partial differential equations and learning operator mappings, respectively. In this regard, a comprehensive study of UQ methods tailored specifically for scientific...

[82] [Autoregressive regularized score-based diffusion models for multi-scenarios fluid flow prediction](#)

Wilfried Genuist, ..., and Didier Clouteau. ArXiv, 2025. 2 citations.

0% Topic Match

Abstract: Building on recent advances in scientific machine learning and generative modeling for computational fluid dynamics, we propose a conditional score-based diffusion model designed for multi-scenarios fluid flow prediction. Our model integrates an energy constraint rooted in the statistical properties of turbulent flows, improving prediction quality with minimal training, while enabling efficient sampling at low cost. The method features a simple and general architecture that requires no problem-specific design, supports plug-and-play enhancements, and enables fast and flexible solution generation. It also demonstrates an efficient conditioning mechanism that simplifies training across different scenarios without demanding a redesign of existing models. We further...

[83] [A selective review of modern stochastic modeling: SDE/SPDE numerics, data-driven identification, and generative methods with applications in biomathematics](#)

Y. Sabbar and K. S. Nisar. Transactions on Computational Modeling and Intelligent Systems, 2025. 5 citations.

0% Topic Match

Abstract: This review maps 2020-2025 developments in stochastic modeling, highlighting non-standard approaches and their applications to biology and epidemiology. It brings together four strands: (1) core models for systems that evolve with randomness; (2) learning key parts of those models directly from data; (3) methods

that can generate realistic synthetic data in continuous time; and (4) numerical techniques that keep simulations stable, accurate, and faithful over long runs. The objective is practical: help researchers quickly see what is new, how the pieces fit together, and where important gaps remain. We summarize tools for estimating changing infection or reaction rates under noisy...

[84] [Learning Hidden Physics and System Parameters with Deep Operator Networks](#)

Vijay Kag, ..., and Somdatta Goswami. ArXiv, 2024. 8 citations.

0% Topic Match

Abstract: Discovering hidden physical laws and identifying governing system parameters from sparse observations are central challenges in computational science and engineering. Existing data-driven methods, such as physics-informed neural networks (PINNs) and sparse regression, are limited by their need for extensive retraining, sensitivity to noise, or inability to generalize across families of partial differential equations (PDEs). In this work, we introduce two complementary frameworks based on deep operator networks (DeepONet) to address these limitations. The first, termed the Deep Hidden Physics Operator (DHPO), extends hidden-physics modeling into the operator-learning paradigm, enabling the discovery of unknown PDE terms across diverse equation families by...

[85] [The ADMM-PINNs Algorithmic Framework for Nonsmooth PDE-Constrained Optimization: A Deep Learning Approach](#)

Yongcun Song, ..., and Hangrui Yue. ArXiv, 2023. 0 citations.

0% Topic Match

Abstract: We study the combination of the alternating direction method of multipliers (ADMM) with physics-informed neural networks (PINNs) for a general class of nonsmooth partial differential equation (PDE)-constrained optimization problems, where additional regularization can be employed for constraints on the control or design variables. The resulting ADMM-PINNs algorithmic framework substantially enlarges the applicable range of PINNs to nonsmooth cases of PDE-constrained optimization problems. The application of the ADMM makes it possible to untie the PDE constraints and the nonsmooth regularization terms for iterations. Accordingly, at each iteration, one of the resulting subproblems is a smooth PDE-constrained optimization which can be efficiently...

[86] [IB-UQ: Information bottleneck based uncertainty quantification for neural function regression and neural operator learning](#)

Ling Guo, ..., and Tao Zhou. ArXiv, 2023. 25 citations.

0% Topic Match

Abstract: We propose a novel framework for uncertainty quantification via information bottleneck (IB-UQ) for scientific machine learning tasks, including deep neural network (DNN) regression and neural operator learning (DeepONet). Specifically, we incorporate the bottleneck by a confidence-aware encoder, which encodes inputs into latent representations according to the confidence of the input data belonging to the region where training data is located, and utilize a Gaussian decoder to predict means and variances of outputs conditional on representation variables. Furthermore, we propose a data augmentation based information bottleneck objective which can enhance the quantification quality of the extrapolation uncertainty, and the encoder and...

[87] [A Vorticity-Enhanced Physics-Informed Neural Network with Logarithmic Reynolds Embedding](#)

Yaxiong Zheng, ..., and Shan Pian. Fluids, 2026. 0 citations.

0% Topic Match

Abstract: To improve unified modeling of steady two-dimensional lid-driven cavity flow across a wide range of Reynolds numbers, this study proposes a Vorticity-Enhanced Physics-Informed Neural Network (VE-PINN). The method augments a standard velocity-pressure PINN with a vorticity-transport residual and uses a logarithmic Reynolds-number embedding, $\log_{10}Re$, for multi-regime training. Using CFD benchmark data as supervision and evaluation, we conduct systematic ablation studies on network architecture, loss weighting, sampling density, input embedding, and physical constraint over $Re=1000$ 50000, together with out-of-range extrapolation tests. The results show that the logarithmic Reynolds-number embedding improves cross-regime training stability and reduces the multi-regime mean relative error, while the...

[88] [Test-time Generalization for Physics through Neural Operator Splitting](#)

Louis Serrano, ..., and Rudy Morel. ArXiv, 2026. 2 citations.

0% Topic Match

Abstract: Neural operators have shown promise in learning solution maps of partial differential equations (PDEs), but they often struggle to generalize when test inputs lie outside the training distribution, such as novel initial conditions, unseen PDE coefficients or unseen physics. Prior works address this limitation with large-scale multiple physics pretraining followed by fine-tuning, but this still requires examples from the new dynamics, falling short of true zero-shot generalization. In this work, we propose a method to enhance generalization at test time, i.e., without modifying pretrained weights. Building on DISCO, which provides a dictionary of neural operators trained across different dynamics, we...

[89] [Physical Information-Based Mach Number Prediction and Model Migration in Continuous Wind Tunnels](#)

Luping Zhao and Chongxiao Wang. Aerospace, 2025. 1 citations.

0% Topic Match

Abstract: In wind tunnel tests for aerospace and bridge engineering, the accurate prediction of Mach number remains a core challenge to ensure the reliability of airflow dynamics characterization. Pure data-driven models often fail to meet high-precision prediction requirements due to the lack of physical mechanism constraints and insufficient generalization capability. This paper proposes a physical information-based long short-term memory network (P-LSTM), which constructs a physical loss function by embedding isentropic flow equations from gas dynamics, thereby constraining the Mach number prediction solution space within the physically feasible domain. This approach effectively balances the neural network's ability to capture temporal features with...

[90] [Physics-Informed Deep Inverse Operator Networks for Solving PDE Inverse Problems](#)

S. Cho and Hwijae Son. ArXiv, 2024. 11 citations.

0% Topic Match

Abstract: Inverse problems involving partial differential equations (PDEs) can be seen as discovering a mapping from measurement data to unknown quantities, often framed within an operator learning approach. However, existing methods typically rely on large amounts of labeled training data, which is impractical for most real-world applications. Moreover, these supervised models may fail to capture the underlying physical principles accurately. To address these limitations, we propose a novel architecture called Physics-Informed Deep Inverse Operator Networks (PI-DIONs), which can learn the solution operator of PDE-based inverse problems without labeled training data. We extend the stability estimates established in the inverse problem literature...

[91] [Uncertainty quantification for noisy inputs-outputs in physics-informed neural networks and neural operators](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2023. 41 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) in scientific machine learning (SciML) becomes increasingly critical as neural networks (NNs) are being widely adopted in addressing complex problems across various scientific disciplines. Representative SciML models are physics-informed neural networks (PINNs) and neural operators (NOs). While UQ in SciML has been increasingly investigated in recent years, very few works have focused on addressing the uncertainty caused by the noisy inputs, such as spatial-temporal coordinates in PINNs and input functions in NOs. The presence of noise in the inputs of the models can pose significantly more challenges compared to noise in the outputs of the models, primarily...

[92] [GSPINN: A novel fast Green kernel solver based on symmetric Physics-Informed neural networks](#)

Xiaopei Jiao and Fansheng Xiong. ArXiv, 2024. 1 citations.

0% Topic Match

No summary or abstract available

[93] [Spatio-Temporal Uncertainty-Modulated Physics-Informed Neural Networks for Solving Hyperbolic Conservation Laws with Strong Shocks](#)

Darui Zhao, ..., and Fujun Liu. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) frequently encounter difficulties in accurately resolving shock waves within high-speed compressible flows, a failure largely attributed to the "gradient pathology" arising from extreme stiffness at discontinuities. To overcome this limitation, we propose the Spatio-Temporal Uncertainty-Modulated PINN (UM-PINN), a probabilistic framework that reinterprets the training process as a multi-task learning problem governed by homoscedastic aleatoric uncertainty. By integrating a gradient-based spatial mask with learnable variance parameters, our method dynamically balances the conflicting contributions of Partial Differential Equation (PDE) residuals and initial conditions across the spatiotemporal domain, further stabilized by Quasi-Monte Carlo Sobol sampling. We validate the framework against challenging...

[94] [Likelihood-Weighted Active Selection of Training Data for Improved Prediction 1 of the Statistics of Extreme Weather Events](#)

B. Champenois and †. Themistoklis P. Sapsis. Journal Not Provided, Unknown year. 0 citations.

0% Topic Match

No summary or abstract available

[95] [Review and Examination of Input Feature Preparation Methods and Machine Learning Models for Turbulence Modeling.](#)

Shirui Luo, ..., and V. Kindratenko. arXiv: Fluid Dynamics, 2020. 3 citations.

0% Topic Match

Abstract: Model extrapolation to unseen flow is one of the biggest challenges facing data-driven turbulence modeling, especially for models with high dimensional inputs that involve many flow features. In this study we review previous efforts on data-driven Reynolds-Averaged Navier Stokes (RANS) turbulence modeling and model extrapolation, with main focus on the popular methods being used in the field of transfer learning. Several potential metrics to measure the dissimilarity between training flows and testing flows are examined. Different Machine Learning (ML) models are compared to understand how the capacity or complexity of the model affects its behavior in the face of dataset...

[96] [Evidential Physics-Informed Neural Networks](#)

H. Tan, ..., and R. McBeth. ArXiv, 2025. 4 citations.

0% Topic Match

Abstract: We present a novel class of Physics-Informed Neural Networks that is formulated based on the principles of Evidential Deep Learning, where the model incorporates uncertainty quantification by learning parameters of a higher-order distribution. The dependent and trainable variables of the PDE residual loss and data-fitting loss terms are recast as functions of the hyperparameters of an evidential prior distribution. Our model is equipped with an information-theoretic regularizer that contains the Kullback-Leibler divergence between two inverse-gamma distributions characterizing predictive uncertainty. Relative to Bayesian-Physics-Informed-Neural-Networks, our framework appeared to exhibit higher sensitivity to data noise, preserve boundary conditions more faithfully and yield empirical...

[97] [E-PINNs: Epistemic Physics-Informed Neural Networks](#)

Ashish S. Nair, ..., and Panos Stinis. ArXiv, 2025. 4 citations.

0% Topic Match

Abstract: Physics-informed neural networks (PINNs) have demonstrated promise as a framework for solving forward and inverse problems involving partial differential equations. Despite recent progress in the field, it remains challenging to quantify uncertainty in these networks. While techniques such as Bayesian PINNs (B-PINNs) provide a principled approach to capturing epistemic uncertainty through Bayesian inference, they can be computationally expensive for large-scale applications. In this work, we propose Epistemic Physics-Informed Neural Networks (E-PINNs), a framework that uses a small network, the epinet, to efficiently quantify epistemic uncertainty in PINNs. The proposed approach works as an add-on to existing, pre-trained PINNs with a...

[98] [Invertible Fourier Neural Operators for Tackling Both Forward and Inverse Problems](#)

Da Long and Shandian Zhe. Unknown journal, 2024. 21 citations.

0% Topic Match

Abstract: Fourier Neural Operator (FNO) is a powerful and popular operator learning method. However, FNO is mainly used in forward prediction, yet a great many applications rely on solving inverse problems. In this paper, we propose an invertible Fourier Neural Operator (iFNO) for jointly tackling the forward and inverse problems. We developed a series of invertible Fourier blocks in the latent channel space to share the model parameters, exchange the information, and mutually regularize the learning for the bi-directional tasks. We integrated a variational auto-encoder to capture the intrinsic structures within the input space and to enable posterior inference so as...

[99] [Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations](#)

M. Raissi, ..., and G. Karniadakis. J. Comput. Phys., 2019. 17081 citations.

0% Topic Match

Abstract: Abstract We introduce physics-informed neural networks – neural networks that are trained to solve supervised learning tasks while respecting any given laws of physics described by general nonlinear partial differential equations. In this work, we present our developments in the context of solving two main classes of problems: data-driven solution and data-driven discovery of partial differential equations. Depending on the nature and arrangement of the available data, we devise two distinct types of algorithms, namely continuous time and discrete time models. The first type of models forms a new family of data-efficient spatio-temporal function approximators, while the latter type allows...

[100] [Uncovering near-wall blood flow from sparse data with physics-informed neural networks](#)

Amirhossein Arzani, ..., and R. D'Souza. Physics of Fluids, 2021. 256 citations.

0% Topic Match

Abstract: Near-wall blood flow and wall shear stress (WSS) regulate major forms of cardiovascular disease, yet they are challenging to quantify with high fidelity. Patient-specific computational and experimental measurement of WSS suffers from uncertainty, low resolution, and noise issues. Physics-informed neural networks (PINN) provide a flexible deep learning framework to integrate mathematical equations governing blood flow with measurement data. By leveraging knowledge about the governing equations (herein, Navier-Stokes), PINN overcomes the large data requirement in deep learning. In this study, it was shown how PINN could be used to improve WSS quantification in diseased arterial flows. Specifically, blood flow problems where...

[101] [Physics-informed machine learning](#)

G. Karniadakis, ..., and Liu Yang. Nature Reviews Physics, 2021. 6721 citations.

0% Topic Match

No summary or abstract available

[102] [\\$PINN - a Domain Decomposition Method for Bayesian Physics-Informed Neural Networks](#)

J'ulia Vicens Figueres, ..., and Khemraj Shukla. ArXiv, 2025. 2 citations.

0% Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) are a novel computational approach for solving partial differential equations (PDEs) with noisy and sparse initial and boundary data. Although, efficient quantification of epistemic and aleatoric uncertainties in big multi-scale problems remains challenging. We propose \$PINN a novel method of computing global uncertainty in PDEs using a Bayesian framework, by combining local Bayesian Physics-Informed Neural Networks (BPINN) with domain decomposition. The solution continuity across subdomains is obtained by imposing the flux continuity across the interface of neighboring subdomains. To demonstrate the effectiveness of \$PINN, we conduct a series of computational experiments on PDEs in 1D and...

[103] [Enabling Local Neural Operators to perform Equation-Free System-Level Analysis](#)

Gianluca Fabiani, ..., and Ioannis G. Kevrekidis. ArXiv, 2025. 14 citations.

0% Topic Match

Abstract: Neural Operators (NOs) provide a powerful framework for computations involving physical laws that can be modelled by (integro-) partial differential equations (PDEs), directly learning maps between infinite-dimensional function spaces that bypass both the explicit equation identification and their subsequent numerical solving. Still, NOs have so far primarily been employed to explore the dynamical behavior as surrogates of brute-force temporal simulations/predictions. Their potential for systematic rigorous numerical system-level tasks, such as fixed-point, stability, and bifurcation analysis - crucial for predicting irreversible transitions in real-world phenomena - remains largely unexplored. Toward this aim, inspired by the Equation-Free multiscale framework, we propose and...

[104] [Understanding and mitigating gradient pathologies in physics-informed neural networks](#)

Sifan Wang, ..., and P. Perdikaris. ArXiv, 2020. 350 citations.

0% Topic Match

Abstract: The widespread use of neural networks across different scientific domains often involves constraining them to satisfy certain symmetries, conservation laws, or other domain knowledge. Such constraints are often imposed as soft penalties during model training and effectively act as domain-specific regularizers of the empirical risk loss. Physics-informed neural networks is an example of this philosophy in which the outputs of deep neural networks are constrained to approximately satisfy a given set of partial differential equations. In this work we review recent advances in scientific machine learning with a specific focus on the effectiveness of physics-informed neural networks in predicting outcomes...

[105] [Physics-Informed Neural Networks for Multiphysics Data Assimilation with Application to Subsurface Transport](#)

Qizhi He, ..., and A. Tartakovsky. ArXiv, 2019. 348 citations.

0% Topic Match

Abstract: Data assimilation for parameter and state estimation in subsurface transport problems remains a significant challenge due to the sparsity of measurements, the heterogeneity of porous media, and the high computational cost of forward numerical models. We present a physics-informed deep neural networks (DNNs) machine learning method for estimating space-dependent hydraulic conductivity, hydraulic head, and concentration fields from sparse measurements. In this approach, we employ individual DNNs to approximate the unknown parameters (e.g., hydraulic conductivity) and states (e.g., hydraulic head and concentration) of a physical system, and jointly train these DNNs by minimizing the loss function that consists of the governing...

[106] [Estimating the carbon footprint of computational fluid dynamics](#)

J. Horwitz. Physics of Fluids, 2024. 9 citations.

0% Topic Match

Abstract: Computational resources have grown exponentially in the past few decades. These machines make possible research and design in fields as diverse as medicine, astronomy, and engineering. Despite ever-increasing computational capabilities, direct simulation of complex systems has remained challenging owing to the degrees of freedom involved. At the cusp of exascale computing, high-resolution simulation of practical problems with minimal model assumptions may soon experience a renaissance. However, growing reliance on modern computers comes at the cost of a growing carbon footprint. To illustrate this, we examine historic computations in fluid dynamics where larger computers have afforded the opportunity to simulate flows...

[107] [Neural-Schwarz Tiling for Geometry-Universal PDE Solving at Scale](#)

P. Secchi, ..., and Marco Maurizi. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Most learned PDE solvers follow a global-surrogate paradigm: a neural operator is trained to map full problem descriptions to full solution fields for a prescribed distribution of geometries, boundary conditions, and coefficients. This has enabled fast inference within fixed problem families, but limits reuse across new domains and makes large-scale deployment dependent on expensive problem-specific data generation. We introduce NEST ($\text{Ne}\text{-}\text{S}\text{chwarz}\text{-}\text{T}\text{iling}$), a local-to-global framework that shifts learning from full-domain solution operators to reusable local physical solvers. The central premise is that, although global PDE solutions depend on geometry, scale, and boundary conditions, the physical response on small neighborhoods can...

[108] [L-HYDRA: Multi-Head Physics-Informed Neural Networks](#)

Zongren Zou and G. Karniadakis. J. Comput. Phys., 2023. 46 citations.

0% Topic Match

Abstract: We introduce multi-head neural networks (MH-NNs) to physics-informed machine learning, which is a type of neural networks (NNs) with all nonlinear hidden layers as the body and multiple linear output layers as multi-head. Hence, we construct multi-head physics-informed neural networks (MH-PINNs) as a potent tool for multi-task learning (MTL), generative modeling, and few-shot learning for diverse problems in scientific machine learning (SciML). MH-PINNs connect multiple functions/tasks via a shared body as the basis functions as well as a shared distribution for the head. The former is accomplished by solving multiple tasks with MH-PINNs with each head independently corresponding to each...

[109] [A Review of Uncertainty Quantification in Deep Learning: Techniques, Applications and Challenges](#)

M. Abdar, ..., and S. Nahavandi. Inf. Fusion, 2020. 2574 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) plays a pivotal role in reduction of uncertainties during both optimization and decision making processes. It can be applied to solve a variety of real-world applications in science and engineering. Bayesian approximation and ensemble learning techniques are two most widely-used UQ methods in the literature. In this regard, researchers have proposed different UQ methods and examined their performance in a variety of applications such as computer vision (e.g., self-driving cars and object detection), image processing (e.g., image restoration), medical image analysis (e.g., medical image classification and segmentation), natural language processing (e.g., text classification, social media texts and...

[110] [Learning to solve Bayesian inverse problems: An amortized variational inference approach](#)

Sharmila Karumuri and Ilias Bilionis. ArXiv, 2023. 6 citations.

0% Topic Match

Abstract: Inverse problems, i.e., estimating parameters of physical models from experimental data, are ubiquitous in science and engineering. The Bayesian formulation is the gold standard because it alleviates ill-posedness issues and quantifies epistemic uncertainty. Since analytical posteriors are not typically available, one resorts to Markov chain Monte Carlo sampling or approximate variational inference. However, inference needs to be rerun from scratch for each new set of data. This drawback limits the applicability of the Bayesian formulation to real-time settings, e.g., health monitoring of engineered systems, and medical diagnosis. The objective of this paper is to develop a methodology that enables real-time...

[111] [Physics-Informed Neural Networks for Nonhomogeneous Material Identification in Elasticity Imaging](#)

Enrui Zhang, ..., and G. Karniadakis. ArXiv, 2020. 78 citations.

0% Topic Match

Abstract: We apply Physics-Informed Neural Networks (PINNs) for solving identification problems of nonhomogeneous materials. We focus on the problem with a background in elasticity imaging, where one seeks to identify the nonhomogeneous mechanical properties of soft tissue based on the full-field displacement measurements under quasi-static loading. In our model, we apply two independent neural networks, one for approximating the solution of the corresponding forward problem, and the other for approximating the unknown material parameter field. As a proof of concept, we validate our model on a prototypical plane strain problem for incompressible hyperelastic tissue. The results show that the PINNs are...

[112] [Learning and discovering multiple solutions using physics-informed neural networks with random initialization and deep ensemble](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2025. 20 citations.

0% Topic Match

Abstract: We explore the capability of physics-informed neural networks (PINNs) to discover multiple solutions. Many real-world phenomena governed by nonlinear differential equations (DEs), such as fluid flow, exhibit multiple solutions under the same conditions, yet capturing this solution multiplicity remains a significant challenge. A key difficulty lies in providing appropriate initial conditions or guesses, as widely used time-marching schemes and Newton's method are highly sensitive to these choices when solving complex computational problems. While machine learning models, particularly PINNs, have shown promise in solving DEs, their ability to capture multiple solutions remains underexplored. In this work, we propose a simple and...

[113] [Property-Driven Testing of Black-Box Functions](#)

Arnab Sharma, ..., and H. Wehrheim. 2022 IEEE/ACM 10th International Conference on Formal Methods in Software Engineering (FormalISE), 2022. 6 citations.

0% Topic Match

Abstract: Testing is one of the most frequent means of quality assurance for software. Property-based testing aims at generating test suites for checking code against user-defined properties. Test input generation is, however, most often independent of the property to be checked, and is instead based on random or user-defined data generation. In this paper, we present property-driven unit testing of functions with numerical inputs and outputs. Unlike property-based testing, it allows users to define the properties to be tested for. Contrary to property-based testing, it also uses the property for a targeted generation of test inputs. Our approach is a form of...

[114] [WLNO: Wavelet-Laplace Neural Operator for Solving Partial Differential Equations](#)

Muhammad Abid, ..., and O. San. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: This work introduces the Wavelet-Laplace Neural Operator (WLNO), a novel neural operator that fuses Haar wavelet multi-scale spatial decomposition with the Laplace-domain pole-residue formulation of the Laplace Neural Operator (LNO). While LNO captures transient and steady-state dynamics through learnable system poles and residues, it lacks an explicit mechanism for extracting spatially localized multi-scale features inherent in complex PDE solutions. WLNO addresses this by augmenting the LNO core with a parallel single-level Haar discrete wavelet transform (DWT) branch that decomposes the lifted feature map into four frequency subbands: approximation (LL), horizontal detail (LH), vertical detail (HL), and diagonal detail (HH) and...

[115] [Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators](#)

Lu Lu, ..., and G. Karniadakis. Nature Machine Intelligence, 2019. 3714 citations.

0% Topic Match

Abstract: It is widely known that neural networks (NNs) are universal approximators of continuous functions. However, a less known but powerful result is that a NN with a single hidden layer can accurately approximate any nonlinear continuous operator. This universal approximation theorem of operators is suggestive of the structure and potential of deep neural networks (DNNs) in learning continuous operators or complex systems from streams of scattered data. Here, we thus extend this theorem to DNNs. We design a new network with small generalization error, the deep operator network (DeepONet), which consists of a DNN for encoding the discrete input function...

[116] [Science and Statistics](#)

G. Box. Journal of the American Statistical Association, 1976. 2190 citations.

0% Topic Match

No summary or abstract available

[117] [Boosting Metamorphic Testing: A General Metamorphic Specification Language and A Supporting System](#)

Chang-ai Sun, ..., and Pak-Lok Poon. ACM Transactions on Software Engineering and Methodology, 2026. 0 citations.

0% Topic Match

Abstract: Metamorphic testing (MT) is a black-box testing technique to alleviate the oracle problem via leveraging Metamorphic Relations (MRs) based on the domain knowledge of the software under test. In recent years, software testing researchers have made significant research advancements on MT in fundamental theories (e.g., MR identification and composition), methodologies (e.g., test input generation), and fault detection effectiveness in various application domains. However, there still exist some major challenges of MT yet to be addressed, such as the needs for a general MR description language, and an automated system that supports all major steps of MT and integrates the various...

[118] [Physics-Informed Neural Operator for Learning Partial Differential Equations](#)

Zong-Yi Li, ..., and Anima Anandkumar. ACM/IMS Journal of Data Science, 2021. 803 citations.

0% Topic Match

Abstract: In this article, we propose physics-informed neural operators (PINO) that combine training data and physics constraints to learn the solution operator of a given family of parametric Partial Differential Equations (PDE). PINO is the first hybrid approach incorporating data and PDE constraints at different resolutions to learn the operator. Specifically, in PINO, we combine coarse-resolution training data with PDE constraints imposed at a higher resolution. The resulting PINO model can accurately approximate the ground-truth solution operator for many popular PDE families and shows no degradation in accuracy even under zero-shot super-resolution, that is, being able to predict beyond the resolution...

[119] [Using Metamorphic Testing to Evaluate DNN Coverage Criteria](#)

Jinyi Zhou, ..., and P. Poon. 2020 IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), 2020. 3 citations.

0% Topic Match

Abstract: Generating test cases and further evaluating their "quality" are two critical topics in the area of Deep Neural Networks (DNNs). In this domain, different studies (e.g., [1], [2]) have reported that metamorphic testing (MT) serves as an effective test case generation method, where an initial set of source test cases is augmented with identified metamorphic relations (MRs) to produce the corresponding set of follow-up test cases. As a result, the fault detection effectiveness (and, hence, the "quality") of the resulting test suite T , containing these source and follow-up test cases, will most likely be increased.

[120] [Scenario Driven Metamorphic Testing for Autonomous Driving Simulators](#)

Yifan Zhang, ..., and Chen Xie. Software Testing, 2024. 6 citations.

0% Topic Match

Abstract: The proliferation of driver assistance features in vehicles has resulted in a growing interest among the public in fully autonomous driving systems (ADSs). However, the integration of software and hardware in these complex systems presents significant testing challenges, particularly with respect to ensuring passenger safety. To address these challenges, simulation has emerged as a crucial step in the testing of ADSs. This paper presents a solution to the challenges faced in testing ADSs, with a focus on the validation of ADS simulators. The proposed approach involves using simulations and metamorphic testing (MT) to generate multiple concrete metamorphic relations (MRs) for testing...

[121] [Inducing Implicit Focus Through Structured Irrelevance for Robust Neural PDE Solvers](#)

Yunpeng Gong, ..., and Min Jiang. 2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND), 2025. 0 citations.

0% Topic Match

Abstract: Neural solvers for partial differential equations (PDEs) have shown remarkable potential across scientific computing and physics-informed learning. However, their performance often hinges on fragile feature dependencies, leading to poor generalization and vulnerability to input noise or domain shifts. In this work, we propose a simple yet effective training-time perturbation scheme that improves generalization by injecting structured, semantically irrelevant signals into the input space. Specifically, we randomly replace small, localized regions of the input with constant values, thereby compelling the model to focus on predictive patterns that remain stable under partial information loss. We formally analyze the learning dynamics and show...

[122] [Metamorphic relations via relaxations: an approach to obtain oracles for action-policy testing](#)

Timo P. Gros and Valentin Wüstholtz. Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis, 2022. 21 citations.

0% Topic Match

Abstract: Testing is a promising way to gain trust in a learned action policy. In particular if π is a neural network. A "bug" in this context constitutes undesirable or fatal policy behavior, e.g., satisfying a failure condition. But how do we distinguish whether such behavior is due to bad policy decisions, or whether it is actually unavoidable under the given circumstances? This requires knowledge about optimal solutions, which defeats the scalability of testing. Related problems occur in software testing when the correct program output is not known. Metamorphic testing addresses this issue through metamorphic relations, specifying how a given...

[123] [Fourier Neural Operator for Plasma Modelling](#)

Vignesh Gopakumar, ..., and Mast Team. ArXiv, 2023. 31 citations.

0% Topic Match

Abstract: Predicting plasma evolution within a Tokamak is crucial to building a sustainable fusion reactor. Whether in the simulation space or within the experimental domain, the capability to forecast the spatio-temporal evolution of plasma field variables rapidly and accurately could improve active control methods on current tokamak devices and future fusion reactors. In this work, we demonstrate the utility of using Fourier Neural Operator (FNO) to model the plasma evolution in simulations and experiments. Our work shows that the FNO is capable of predicting magnetohydrodynamic models governing the plasma dynamics, 6 orders of magnitude faster than the traditional numerical solver, while...

[124] [Metamorphic relation automation: Rationale, challenges, and solution directions](#)

Emran Altamimi, ..., and C. Catal. Journal of Software: Evolution and Process, 2022. 4 citations.

0% Topic Match

Abstract: Metamorphic testing addresses the issue of the oracle problem by comparing results transformation from multiple test executions. The relationship that governs the output transformation is called metamorphic relation. Metamorphic relations require expert knowledge and the generation of them is considered a

time consuming task. Researchers have proposed various techniques to automate metamorphic testing, generation, and selection. Although there are several research articles on this issue, there is a lack of overview of the state of the art of metamorphic relation automation. As such, we performed a systematic literature review study to collect, extract, and synthesize the required data. Based on our research questions, the...

[125] [Adaptive Metamorphic Testing for Object Detection Systems](#)

Ruifeng Li, ..., and Zheng Zheng. 2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C), 2024. 1 citations.

0% Topic Match

Abstract: Metamorphic testing effectively mitigates the oracle problem, consequently gaining extensive utilization in the testing of machine learning systems. Present research on metamorphic testing is predominantly focused on the design and application of metamorphic relations, with scant attention to the selection and parameter configuration of metamorphic relations during the testing process to expedite fault detection. This paper presents a novel testing methodology for deep learning object detection systems, named ADRMT for Adaptive Dynamic Random Metamorphic Testing, which leverages metamorphic testing alongside dynamic random testing and adaptive random testing. By harnessing the principles of dynamic random testing to guide the selection of...

[126] [Metamorphic Testing of Vision-Language Action-Enabled Robots](#)

Pablo Valle, ..., and Aitor Arrieta. ArXiv, 2026. 1 citations.

0% Topic Match

Abstract: Vision-Language-Action (VLA) models are multimodal robotic task controllers that, given an instruction and visual inputs, produce a sequence of low-level control actions (or motor commands) enabling a robot to execute the requested task in the physical environment. These systems face the test oracle problem from multiple perspectives. On the one hand, a test oracle must be defined for each instruction prompt, which is a complex and non-generalizable approach. On the other hand, current state-of-the-art oracles typically capture symbolic representations of the world (e.g., robot and object states), enabling the correctness evaluation of a task, but fail to assess other critical...

[127] [Deep neural operators can serve as accurate surrogates for shape optimization: A case study for airfoils](#)

K. Shukla, ..., and G. Karniadakis. ArXiv, 2023. 23 citations.

0% Topic Match

Abstract: Deep neural operators, such as DeepONets, have changed the paradigm in high-dimensional nonlinear regression from function regression to (differential) operator regression, paving the way for significant changes in computational engineering applications. Here, we investigate the use of DeepONets to infer flow fields around unseen airfoils with the aim of shape optimization, an important design problem in aerodynamics that typically taxes computational resources heavily. We present results which display little to no degradation in prediction accuracy, while reducing the online optimization cost by orders of magnitude. We consider NACA airfoils as a test case for our proposed approach, as their shape...

[128] [Fast Reconstruction of Exact Maxwell Dynamics from Sparse Data](#)

Dan DeGenaro, ..., and Bogdan Raictua. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: We introduce FLASH-MAX, a shallow, exact-by-construction neural network architecture for predicting homogeneous electromagnetic fields from sparse pointwise observations. Each hidden neuron represents a separate exact solution to Maxwell's equations, so that the network satisfies the governing equations symbolically by construction and can be trained end-to-end from sparse data within seconds. We prove a universal approximation result showing that this exact model class remains universal on arbitrary domains. FLASH-MAX reaches sub-1% relative validation error from about 1K sparse pointwise observations in seconds, all while maintaining a zero PDE residual, and keeps single-digit errors even for only 100 observations sampled from 3D...

[129] [Engineering Applications of Artificial Intelligence](#)

K. Shukla, ..., and O. Articleinf. Journal Not Provided, Unknown year. 0 citations.

0% Topic Match

No summary or abstract available

[130] [A Dynamic Recognition Method of Metamorphic Relation Identification](#)

Jie Hong, ..., and Helin Gong. 2022 13th International Conference on Reliability, Maintainability, and Safety (ICRMS), 2022. 4 citations.

0% Topic Match

Abstract: Metamorphic testing is one of the effective methods to alleviate the test oracle problem. Metamorphic relation is the core of metamorphic testing, and there is no effective automatic identification technology. This paper transforms the metamorphic relation recognition issue into a symbolic expression regression problem. The test input pairs are generated using the preset input pattern, and the output pattern is mined by gene expression programming. After reduction and verification, a valid metamorphic relation is obtained. Experiments show that this method can identify the relation obtained by static analysis and a more significant number. At the same time, compared with existing...

[131] [Basic Framework and Main Methods of Uncertainty Quantification](#)

Juan Zhang, ..., and Ruili Wang. Mathematical Problems in Engineering, 2020. 63 citations.

0% Topic Match

Abstract: Since 2000, the research of uncertainty quantification (UQ) has been successfully applied in many fields and has been highly valued and strongly supported by academia and industry. This review firstly discusses the sources and the types of uncertainties and gives an overall discussion on the goal, practical significance, and basic framework of the research of UQ. Then, the core ideas and typical methods of several important UQ processes are introduced, including sensitivity analysis, uncertainty propagation, model calibration, Bayesian inference, experimental design, surrogate model, and model uncertainty analysis.

[132] [Metamorphic testing of large language model question answering systems](#)

Xiaoyin Wang and Xinyu Liu. Unknown journal, 2025. 0 citations.

0% Topic Match

Abstract: Large language models are capable of producing a wide variety of creative outputs, and their question-answering capabilities are compelling. Current testing efforts for large language models question answering systems are largely based on the reference paradigm and rely on manual labeling of test inputs. Testing of Large language models suffers from the test oracle problem. Metamorphic testing can test software by defining the expected relationships between inputs and outputs to ensure that the software still works correctly in different situations. In this paper, we propose a metamorphic testing method for large language models question answering systems. Specifically, starting from the...

[133] [Physics-Informed Latent Neural Operator for Real-time Predictions of Complex Physical Systems](#)

Sharmila Karumuri, ..., and Somdatta Goswami. ArXiv, 2025. 8 citations.

0% Topic Match

No summary or abstract available

[134] [Using Metamorphic Relations to Improve The Effectiveness of Automatically Generated Test Cases](#)

Prashanta Saha and Upulee Kanewala. 2022 IEEE/ACIS 20th International Conference on Software Engineering Research, Management and Applications (SERA), 2022. 0 citations.

0% Topic Match

Abstract: Automated test case generation has helped to reduce the cost of testing. However, developing effective test oracles for these automatically generated test cases still remains a challenge. Metamorphic testing (MT) has become a well-known software testing approach over the years. This testing technique can effectively alleviate the oracle problem faced when testing using metamorphic relations (MRs) to determine whether a test case is passed or failed. In this work, we conduct an empirical study on an open source linear algebra library to evaluate whether MRs can be utilized to improve the fault detection effectiveness of automatically generated test cases. Our...

[135] [A Case Study of Metamorphic Test for Monte Carlo Numerical Calculation Programs](#)

Yi Chen, ..., and Xiaohua Yang. 2023 International Conference on Applied Physics and Computing (ICAPC), 2023. 1 citations.

0% Topic Match

Abstract: The Monte Carlo numerical calculation program is based on non-deterministic theory, which encounters the 'test oracle' problem. Metamorphic Test (MT) is an effective method to alleviate this problem. This paper applies MT to test the Monte Carlo numerical calculation programs and proposes a new method for deriving the Metamorphic Relations (MRs) in MT. Firstly, MRs are analyzed through the random sampling model of under-test programs. Secondly, MT is performed. Finally, the results and the effectiveness of MT are analysed. The experiments show that MRs can effectively detect errors in the program, and the proposed method can alleviate the 'test oracle'...

[136] [Leveraging viscous Hamilton-Jacobi PDEs for uncertainty quantification in scientific machine learning](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2024. 13 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) in scientific machine learning (SciML) combines the powerful predictive power of SciML with methods for quantifying the reliability of the learned models. However, two major challenges remain: limited interpretability and expensive training procedures. We provide a new interpretation for UQ problems by establishing a new theoretical connection between some Bayesian inference problems arising in SciML and viscous Hamilton-Jacobi partial differential equations (HJ PDEs). Namely, we show that the posterior mean and covariance can be recovered from the spatial gradient and Hessian of the solution to a viscous HJ PDE. As a first exploration of this connection, we...

[137] [Towards the Automation of Metamorphic Testing in Model Transformations](#)

J. Troya, ..., and Antonio Ruiz-Cortés. ArXiv, 2018. 2 citations.

0% Topic Match

Abstract: Model transformations are the cornerstone of Model-Driven Engineering, and provide the essential mechanisms for manipulating and transforming models. Checking whether the output of a model transformation is correct is a manual and error-prone task, this is referred to as the oracle problem in the software testing literature. The correctness of the model transformation program is crucial for the proper generation of its output, so it should be tested. Metamorphic testing is a testing technique to alleviate the oracle problem consisting on exploiting the relations between different inputs and outputs of the program under test, so-called metamorphic relations. In this paper...

[138] [A SURVEY OF VALIDATION IN HEALTH CARE SIMULATION STUDIES](#)

M. Raunak and Megan M. Olsen. Journal Not Provided, 2014. 7 citations.

0% Topic Match

No summary or abstract available

[139] [Bidirectional Empowerment of Metamorphic Testing and Large Language Models: A Systematic Survey](#)

Zheng Zheng, ..., and Tsong Yueh Chen. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Large language models (LLMs) have introduced substantial challenges to software quality assurance due to their generative, probabilistic, and open-ended nature, which intensifies the oracle problem and limits the applicability of traditional testing methods. Metamorphic testing (MT), which checks necessary relations among multiple related executions rather than relying on exact expected outputs, has emerged as a promising approach for testing LLMs and other oracle-deficient systems. At the same time, the strong semantic understanding, reasoning, and code generation capabilities of LLMs create new opportunities to automate the traditionally labor-intensive phases of MT. This survey systematically reviews 93 primary studies and characterizes this...

[140] [A survey on machine learning approaches for uncertainty quantification of engineering systems](#)

Yan Shi, ..., and Michael Beer. Machine Learning for Computational Science and Engineering, 2025. 57 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) is essential for understanding and mitigating the impact of pervasive uncertainties in engineering systems, playing a crucial role in modern engineering practice. As engineering products grow increasingly complex and the demand for highly accurate UQ results intensifies, the need for efficient UQ methods has become paramount. Recently, machine learning (ML) techniques, including Gaussian process regression, artificial neural networks, physics-informed neural networks, and many others, have garnered significant attention in both theoretical research and practical applications. The exceptional capability of ML methods to address highly complex problems has positioned them as leading solutions for UQ challenges. This work...

[141] [Metamorphic Testing: A New Approach for Generating Next Test Cases](#)

T. Chen, ..., and S. Yiu. ArXiv, 2020. 645 citations.

Not measured Topic Match

Abstract: In software testing, a set of test cases is constructed according to some predefined selection criteria. The software is then examined against these test cases. Three interesting observations have been made on the current artifacts of software testing. Firstly, an error-revealing test case is considered useful while a successful test case which does not reveal software errors is usually not further investigated. Whether these successful test cases still contain useful information for revealing software errors has not been properly studied. Secondly, no matter how extensive the testing has been conducted in the development phase, errors may still exist in the...

[142] [Out-of-distribution generalization of deep-learning surrogates for 2D PDE-generated dynamics in the small-data regime](#)

Binh Duong Nguyen and Stefan Sandfeld. ArXiv, 2026. 2 citations.

Not measured Topic Match

Abstract: Partial differential equations (PDEs) are a central tool for modeling the dynamics of physical, engineering, and materials systems, but high-fidelity simulations are often computationally expensive. At the same time, many scientific applications can be viewed as the evolution of spatially distributed fields, making data-driven forecasting of such fields a core task in scientific machine learning. In this work we study autoregressive deep-learning surrogates for two-dimensional PDE dynamics on periodic domains, focusing on generalization to out-of-distribution initial conditions within a fixed PDE and parameter regime and on strict small-data settings with at most 10^2 simulated trajectories per system. We introduce a...

[143] [Learning Mesh-Based Simulation with Graph Networks](#)

T. Pfaff, ..., and P. Battaglia. ArXiv, 2020. 1242 citations.

Not measured Topic Match

Abstract: Mesh-based simulations are central to modeling complex physical systems in many disciplines across science and engineering. Mesh representations support powerful numerical integration methods and their resolution can be adapted to strike favorable trade-offs between accuracy and efficiency. However, high-dimensional scientific simulations are very expensive to run, and solvers and parameters must often be tuned individually to each system studied. Here we introduce MeshGraphNets, a framework for learning mesh-based simulations using graph neural networks. Our model can be trained to pass messages on a mesh graph and to adapt the mesh discretization during forward simulation. Our results show it can accurately...

[144] [Testing and Validating Machine Learning Classifiers by Metamorphic Testing](#)

Xiaoyuan Xie, ..., and T. Chen. The Journal of systems and software, 2011. 397 citations.

Not measured Topic Match

Abstract: Machine Learning algorithms have provided core functionality to many application domains - such as bioinformatics, computational linguistics, etc. However, it is difficult to detect faults in such applications because often there is no "test oracle" to verify the correctness of the computed outputs. To help address the software quality, in this paper we present a technique for testing the implementations of machine learning classification algorithms which support such applications. Our approach is based on the technique "metamorphic testing", which has been shown to be effective to alleviate the oracle problem. Also presented include a case study on a real-world machine...

[145] [Metamorphic testing of programs on partial differential equations: a case study](#)

T. Chen, ..., and T. H. Tse. Proceedings 26th Annual International Computer Software and Applications, 2002. 123 citations.

Not measured Topic Match

No summary or abstract available

[146] [Embedding machine-learned sub-grid variability improves climate model precipitation patterns](#)

Daniel Giles, ..., and S. Guillas. Communications Earth & Environment, 2024. 5 citations.

Not measured Topic Match

No summary or abstract available

[147] [Validating a Deep Learning Framework by Metamorphic Testing](#)

Junhua Ding, ..., and Xin-Hua Hu. 2017 IEEE/ACM 2nd International Workshop on Metamorphic Testing (MET), 2017. 75 citations.

Not measured Topic Match

No summary or abstract available

[148] [Identifying implementation bugs in machine learning based image classifiers using metamorphic testing](#)

Anurag Dwarakanath, ..., and Sanjay Podder. Proceedings of the 27th ACM SIGSOFT International Symposium on Software Testing and Analysis, 2018. 194 citations.

Not measured Topic Match

Abstract: We have recently witnessed tremendous success of Machine Learning (ML) in practical applications. Computer vision, speech recognition and language translation have all seen a near human level performance. We expect, in the near future, most business applications will have some form of ML. However, testing such applications is extremely challenging and would be very expensive if we follow today's methodologies. In this work, we present an articulation of the challenges in testing ML based applications. We then present our solution approach, based on the concept of Metamorphic Testing, which aims to identify implementation bugs in ML based image classifiers. We...

[149] [The evaluation of program-based software test data adequacy criteria](#)

E. Weyuker. Communications of the ACM, 1988. 169 citations.

Not measured Topic Match

No summary or abstract available

[150] [Plasma surrogate modelling using Fourier neural operators](#)

Vignesh Gopakumar, ..., and Mast Team. Nuclear Fusion, 2023. 53 citations.

Not measured Topic Match

Abstract: Predicting plasma evolution within a Tokamak reactor is crucial to realizing the goal of sustainable fusion. Capabilities in forecasting the spatio-temporal evolution of plasma rapidly and accurately allow us to quickly iterate over design and control strategies on current Tokamak devices and future reactors. Modelling plasma evolution using numerical solvers is often expensive, consuming many hours on supercomputers, and hence, we need alternative inexpensive surrogate models. We demonstrate accurate predictions of plasma evolution both in simulation and experimental domains using deep learning-based surrogate modelling tools, viz., Fourier neural operators (FNO). We show that FNO has a speedup of six orders...

[151] [Active Learning for Neural PDE Solvers](#)

Daniel Musekamp, ..., and Mathias Niepert. ArXiv, 2024. 23 citations.

Not measured Topic Match

Abstract: Solving partial differential equations (PDEs) is a fundamental problem in science and engineering. While neural PDE solvers can be more efficient than established numerical solvers, they often require large amounts of training data that is costly to obtain. Active learning (AL) could help surrogate models reach the same accuracy with smaller training sets by querying classical solvers with more informative initial conditions and PDE parameters. While AL is more common in other domains, it has yet to be studied extensively for neural PDE solvers. To bridge this gap, we introduce AL4PDE, a modular and extensible active learning benchmark. It provides...

[152] [Data efficiency and long term prediction capabilities for neural operator surrogate models of core and edge plasma codes](#)

N. Carey, ..., and J. Brandstetter. ArXiv, 2024. 9 citations.

Not measured Topic Match

Abstract: Simulation-based plasma scenario development, optimization and control are crucial elements towards the successful deployment of next-generation experimental tokamaks and Fusion power plants. Current simulation codes require extremely intensive use of HPC resources that make them unsuitable for iterative or real time applications. Neural network based surrogate models of expensive simulators have been proposed to speed up such costly workflows. Current efforts in this direction in the Fusion community are mostly limited to point estimates of quantities of interest or simple 1D PDE models, with a few notable exceptions. While the AI literature on methods for neural PDE surrogate models is...

[153] [Neural general circulation models for weather and climate](#)

Dmitrii Kochkov, ..., and Stephan Hoyer. Nature, 2023. 472 citations.

Not measured Topic Match

Abstract: General circulation models (GCMs) are the foundation of weather and climate prediction^{1,2}. GCMs are physics-based simulators that combine a numerical solver for large-scale dynamics with tuned representations for small-scale processes such as cloud formation. Recently, machine-learning models trained on reanalysis data have achieved comparable or better skill than GCMs for deterministic weather forecasting^{3,4}. However, these models have not demonstrated improved ensemble forecasts, or shown sufficient stability for long-term weather and climate simulations. Here we present a GCM that combines a differentiable solver for atmospheric dynamics with machine-learning components and show that it can generate forecasts of deterministic weather, ensemble weather...

[154] [Conformal Methods for Quantifying Uncertainty in Spatiotemporal Data: A Survey](#)

S. Sun. ArXiv, 2022. 16 citations.

Not measured Topic Match

Abstract: Machine learning methods are increasingly widely used in high-risk settings such as healthcare, transportation, and finance. In these settings, it is important that a model produces calibrated uncertainty to reflect its own confidence and avoid failures. In this paper we survey recent works on uncertainty quantification (UQ) for deep learning, in particular distribution-free Conformal Prediction method for its mathematical properties and wide applicability. We will cover the theoretical guarantees of conformal methods, introduce techniques that improve calibration and efficiency for UQ in the context of spatiotemporal data, and discuss the role of UQ in the context of safe decision making.

[155] [Fast dynamic 1D simulation of divertor plasmas with neural PDE surrogates](#)

Y. Poels, ..., and V. Menkovski. Nuclear Fusion, 2023. 23 citations.

Not measured Topic Match

Abstract: Managing divertor plasmas is crucial for operating reactor scale tokamak devices due to heat and particle flux constraints on the divertor target. Simulation is an important tool to understand and control these plasmas, however, for real-time applications or exhaustive parameter scans only simple approximations are currently fast enough. We address this lack of fast simulators using neural partial differential equation (PDE) surrogates, data-driven neural network-based surrogate models trained using solutions generated with a classical numerical method. The surrogate approximates a time-stepping operator that evolves the full spatial solution of a reference physics-based model over time. We use DIV1D, a 1D...

[156] [Towards Stability of Autoregressive Neural Operators](#)

Michael McCabe, ..., and Jed Brown. ArXiv, 2023. 57 citations.

Not measured Topic Match

Abstract: Neural operators have proven to be a promising approach for modeling spatiotemporal systems in the physical sciences. However, training these models for large systems can be quite challenging as they incur significant computational and memory expense -- these systems are often forced to rely on autoregressive time-stepping of the neural network to predict future temporal states. While this is effective in managing costs, it can lead to uncontrolled error growth over time and eventual instability. We analyze the sources of this autoregressive error growth using prototypical neural operator models for physical systems and explore ways to mitigate it. We introduce...

[157] [Spatio-Temporal Wildfire Prediction Using Multi-Modal Data](#)

Chen Xu, ..., and Feng Qiu. IEEE Journal on Selected Areas in Information Theory, 2022. 10 citations.

Not measured Topic Match

Abstract: Due to severe societal and environmental impacts, wildfire prediction using multi-modal sensing data has become a highly sought-after data-analytical tool by various stakeholders (such as state governments and power utility companies) to achieve a more informed understanding of wildfire activities and plan preventive measures. A desirable algorithm should precisely predict fire risk and magnitude for a location in real time. In this paper, we develop a flexible spatio-temporal wildfire

prediction framework using multi-modal time series data. We first predict the wildfire risk (the chance of a wildfire event) in real-time, considering the historical events using discrete mutually exciting point process...

[158] [Loss Landscape Engineering via Data Regulation on PINNs](#)

Vignesh Gopakumar, ..., and D. Samaddar. ArXiv, 2022. 30 citations.

Not measured Topic Match

Abstract: Physics-Informed Neural Networks have shown unique utility in parameterising the solution of a well-defined partial differential equation using automatic differentiation and residual losses. Though they provide theoretical guarantees of convergence, in practice the required training regimes tend to be exacting and demanding. Through the course of this paper, we take a deep dive into understanding the loss landscapes associated with a PINN and how that offers some insight as to why PINNs are fundamentally hard to optimise for. We demonstrate how PINNs can be forced to converge better towards the solution, by way of feeding in sparse or coarse data...

[159] [Fourier Neural Operator for Parametric Partial Differential Equations](#)

Zong-Yi Li, ..., and Anima Anandkumar. ArXiv, 2020. 4131 citations.

Not measured Topic Match

Abstract: The classical development of neural networks has primarily focused on learning mappings between finite-dimensional Euclidean spaces. Recently, this has been generalized to neural operators that learn mappings between function spaces. For partial differential equations (PDEs), neural operators directly learn the mapping from any functional parametric dependence to the solution. Thus, they learn an entire family of PDEs, in contrast to classical methods which solve one instance of the equation. In this work, we formulate a new neural operator by parameterizing the integral kernel directly in Fourier space, allowing for an expressive and efficient architecture. We perform experiments on Burgers' equation,...

[160] [MESHFREEFLOWNET: A Physics-Constrained Deep Continuous Space-Time Super-Resolution Framework](#)

C. Jiang, ..., and Anima Anandkumar. SC20: International Conference for High Performance Computing, Networking, Storage and Analysis, 2020. 166 citations.

Not measured Topic Match

Abstract: We propose MESHFREEFLOWNET, a novel deep learning-based super-resolution framework to generate continuous (grid-free) spatio-temporal solutions from the lowresolution inputs. While being computationally efficient, MESHFREEFLOWNET accurately recovers the fine-scale quantities of interest. MESHFREEFLOWNET allows for: (i) the output to be sampled at all spatio-temporal resolutions, (ii) a set of Partial Differential Equation (PDE) constraints to be imposed, and (iii) training on fixed-size inputs on arbitrarily sized spatio-temporal domains owing to its fully convolutional encoder. We empirically study the performance of MESHFREEFLOWNET on the task of super-resolution of turbulent flows in the Rayleigh-Bénard convection problem. Across a diverse set of evaluation metrics,...

[161] [Conformal prediction interval for dynamic time-series](#)

Chen Xu and Yao Xie. Unknown journal, 2020. 188 citations.

Not measured Topic Match

No summary or abstract available

[162] [Machine Learning Emulation of 3D Cloud Radiative Effects](#)

David Meyer, ..., and Shannon L. Mason. Journal of Advances in Modeling Earth Systems, 2021. 34 citations.

Not measured Topic Match

Abstract: The treatment of cloud structure in numerical weather and climate models is often greatly simplified to make them computationally affordable. Here we propose to correct the European Centre for Medium Range Weather Forecasts 1D radiation scheme ecRad for 3D cloud effects using computationally cheap neural networks. 3D cloud effects are learned as the difference between ecRad's fast 1D Tripleclouds solver that neglects them and its 3D SPARTACUS (SPeedy Algorithm for Radiative TrAnsfer through CloUd Sides) solver that includes them but is about five times more computationally expensive. With typical errors between 20% and 30% of the 3D signal, neural networks improve...

[163] [Accelerating the BSM interpretation of LHC data with machine learning](#)

G. Bertone, ..., and M. Welling. Physics of the Dark Universe, 2016. 20 citations.

Not measured Topic Match

Abstract: The interpretation of Large Hadron Collider (LHC) data in the framework of Beyond the Standard Model (BSM) theories is hampered by the need to run computationally expensive event generators and detector simulators. Performing statistically convergent scans of high-dimensional BSM theories is consequently challenging, and in practice unfeasible for very high-dimensional BSM theories. We present here a new machine learning method that accelerates the interpretation of LHC data, by learning the relationship between BSM theory parameters and data. As a proof-of-concept, we demonstrate that this technique accurately predicts natural SUSY signal events in two signal regions at the High Luminosity LHC,...

[164] [Enabling Automatic Differentiation with Mollified Graph Neural Operators](#)

Ryan Lin, ..., and Anima Anandkumar. Trans. Mach. Learn. Res., 2025. 7 citations.

Not measured Topic Match

Abstract: Physics-informed neural operators offer a powerful framework for learning solution operators of partial differential equations (PDEs) by combining data and physics losses. However, these physics losses rely on derivatives. Computing these derivatives remains challenging, with spectral and finite difference methods introducing approximation errors due to finite resolution. Here, we propose the mollified graph neural operator (mGNO), the first method to leverage automatic differentiation and compute exact gradients on arbitrary geometries. This enhancement enables efficient training on irregular grids and varying geometries while allowing seamless evaluation of physics losses at randomly sampled points for improved generalization. For a PDE example on...

[165] [Towards Enforcing Hard Physics Constraints in Operator Learning Frameworks](#)

Valentin Duruisseau, ..., and Anima Anandkumar. Journal Not Provided, Unknown year. 9 citations.

Not measured Topic Match

No summary or abstract available

[166] [Numerical Study of Neural Network Surrogate Model Performance for Uncertainty Propagation](#)

Noah Wade and K. Teferra. Journal of Engineering Mechanics, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

[168] [Physics-informed neural operator for forecasting vehicle-induced bridge vibration trajectories](#)

Rong Chen, ..., and Jiaji Wang. Engineering Applications of Artificial Intelligence, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

[169] [A multi-scale physics-informed convolutional framework with adaptive loss design for parametric steady-state PDEs](#)

Xiaofeng Xie. Engineering Analysis with Boundary Elements, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

[170] [Extended parametric physics-informed neural networks \(ePINNs\) for efficient solution of heat transfer and fluid flow problems](#)

Tongsheng Wang, ..., and Zhu Huang. Case Studies in Thermal Engineering, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

[171] [Neural Quantum Spectral Operator Learning for Solving Partial Differential Equations](#)

Chanyoung Kim, ..., and Youngjoon Hong. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Partial differential equations (PDEs) are central to modeling physical and engineering systems, but repeatedly solving parametric PDEs remains computationally expensive. Operator learning enables fast surrogate inference, yet typically requires large input-output paired datasets generated by costly high-fidelity PDE solvers. Unsupervised operator learning frameworks alleviate data dependency but remain hindered by computational bottlenecks. To address this, we propose Neural Variational Quantum Linear Solver (NVQLS), the first hybrid quantum-classical operator learning framework leveraging the Legendre--Galerkin weak formulation. We critically resolve the sign ambiguity in VQLS energy minimization, preventing erroneous solution representations. Additionally, we introduce a neural embedding, a novel encoding scheme to...

- [172] [Physics-informed Machine Learning for Prognostics and Health Management: Foundations, Advances and Prospects](#)

Danning Sun, ..., and Yupeng Wei. Digital Engineering, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

- [173] [Size Lowerbounds for Deep Operator Networks](#)

Anirbit Mukherjee and Amartya Roy. Trans. Mach. Learn. Res., 2023. 7 citations.

Not measured Topic Match

Abstract: Deep Operator Networks are an increasingly popular paradigm for solving regression in infinite dimensions and hence solve families of PDEs in one shot. In this work, we aim to establish a first-of-its-kind data-dependent lowerbound on the size of DeepONets required for them to be able to reduce empirical error on noisy data. In particular, we show that for low training errors to be obtained on n data points it is necessary that the common output dimension of the branch and the trunk net be scaling as $\Omega(\sqrt[n]{n})$. This inspires our experiments with DeepONets solving the...

- [174] [IV-Net: A neural network for elliptic PDEs with random and highly varying coefficients](#)

Shanni Zhong and G. Birros. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: We introduce a novel neural operator architecture designed to approximate solutions of linear elliptic partial differential equations with high-contrast, spatially varying coefficients. The network, termed the Iterated V-shaped Net (IV-Net), realizes a mapping from the input coefficients and righthand side to the corresponding solution field. The architecture of IV-Net is informed by, and closely resembles, a V-cycle multigrid solver. The IV-Net model is parameterized via convolutional layers defined in the physical domain. For coercive problems with highly heterogeneous coefficients, the proposed network exhibits superior performance relative to a proper orthogonal decomposition (POD) approach and several existing neural operator architectures. For...

- [175] [DeepPropNet: an operator learning-based predictor for thermal plasma properties](#)

Zuo Wang and Linlin Zhong. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Thermal plasma properties play a critical role in plasma simulations and plasma-related applications. However, their strong nonlinear dependence on temperature, pressure, and gas composition makes accurate and efficient evaluation challenging. In this work, an operator learning-based model, termed DeepPropNet, is proposed for fast prediction of thermodynamic and transport properties of thermal plasmas. Two architectures are developed, including a single-property model (S-DeepPropNet) and a Mixture of Experts (MoE)-based multi-property model (MoE-DeepPropNet). The proposed models learn the nonlinear mapping from plasma operating conditions to physical properties based on high-fidelity datasets. The MoE architecture enables efficient multi-property prediction within a unified framework. Predictions...

- [176] [Structure-Aware Epistemic Uncertainty Quantification for Neural Operator PDE Surrogates](#)

Haoze Song, ..., and Wei Wang. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Neural operators (NOs) provide fast, resolution-invariant surrogates for mapping input fields to PDE solution fields, but their predictions can exhibit significant epistemic uncertainty due to finite data, imperfect optimization, and distribution shift. For practical deployment in scientific computing, uncertainty quantification (UQ) must be both computationally efficient and spatially faithful, i.e., uncertainty bands should align with the localized residual structures that matter for downstream risk management. We propose a structure-aware epistemic UQ scheme that exploits the modular anatomy common to modern NOs (lifting-propagation-recovering). Instead of applying unstructured weight perturbations (e.g., naive dropout) across the entire network, we restrict Monte Carlo sampling...

- [177] [Beamforming for Massive MIMO Aerial Communications: A Robust and Scalable DRL Approach](#)

Hesam Khoshkbari, ..., and H. Yanikomeroglu. IEEE Transactions on Communications, 2025. 3 citations.

Not measured Topic Match

Abstract: This paper presents a distributed beamforming framework for a constellation of airborne platform stations (APSSs) in a massive Multiple-Input and Multiple-Output (MIMO) non-terrestrial network (NTN) that targets the downlink sum-rate maximization under imperfect local channel state information (CSI). We propose a novel entropy-based multi-agent deep reinforcement learning (DRL) approach where each non-terrestrial base station (NTBS) independently computes its beamforming vector using a Fourier Neural Operator (FNO) to capture long-range dependencies in the frequency domain. To ensure scalability and robustness, the proposed framework integrates transfer learning based on a conjugate prior mechanism and a low-rank decomposition (LRD) technique, thus enabling efficient...

- [178] [Electron neural closure for turbulent magnetosheath simulations: energy channels](#)

G. Miloshevich, ..., and G. Lapenta. ArXiv, 2025. 2 citations.

Not measured Topic Match

Abstract: In this work, we introduce a non-local five-moment electron pressure tensor closure parametrized by a fully convolutional neural network (FCNN). Electron pressure plays an important role in generalized Ohm's law, competing with electron inertia. This model is used in the development of a surrogate model for a fully kinetic energy-conserving semi-implicit particle-in-cell simulation of decaying magnetosheath turbulence. We achieve this by training FCNN on a representative set of simulations with a smaller number of particles per cell and showing that our results generalize to a simulation with a large number of particles per cell. We evaluate the statistical properties of...

- [179] [Spatiotemporal forecasting of the edge localized modes in tokamak plasmas using neural networks](#)

A. Samaddar, ..., and A. Nelson. Machine Learning: Science and Technology, 2025. 0 citations.

Not measured Topic Match

Abstract: Artificial intelligence techniques have been increasingly adopted by the plasma and fusion science to address problems like plasma reconstruction, surrogate modeling, and tokamak/stellarator optimization. A key focus in sustained fusion research is the prediction and mitigation of edge-localized-modes (ELMs), instabilities that occur in short, periodic bursts and can cause erosion to the tokamak vessel wall. Recent research has demonstrated the power of neural networks in approximating continuous functions. In this work, we build spatiotemporal forecasting models that can predict the onset of ELMs and their evolution at early stages. We leverage recent advances in generative modeling, sequence-to-sequence modeling, and Fourier...

- [180] [Scale-Consistent Learning for Partial Differential Equations](#)

Zong-Yi Li, ..., and Anima Anandkumar. ArXiv, 2025. 2 citations.

Not measured Topic Match

Abstract: Machine learning (ML) models have emerged as a promising approach for solving partial differential equations (PDEs) in science and engineering. Previous ML models typically cannot generalize outside the training data; for example, a trained ML model for the Navier-Stokes equations only works for a fixed Reynolds number (Re) on a pre-defined domain. To overcome these limitations, we propose a data augmentation scheme based on scale-consistency properties of PDEs and design a scale-informed neural operator that can model a wide range of scales. Our formulation leverages the facts: (i) PDEs can be rescaled, or more concretely, a given domain can be...

- [181] [NOBLE - Neural Operator with Biologically-informed Latent Embeddings to Capture Experimental Variability in Biological Neuron Models](#)

Luca Ghafourpour, ..., and Anima Anandkumar. ArXiv, 2025. 4 citations.

Not measured Topic Match

Abstract: Characterizing the cellular properties of neurons is fundamental to understanding their function in the brain. In this quest, the generation of bio-realistic models is central towards integrating multimodal cellular data sets and establishing causal relationships. However, current modeling approaches remain constrained

by the limited availability and intrinsic variability of experimental neuronal data. The deterministic formalism of bio-realistic models currently precludes accounting for the natural variability observed experimentally. While deep learning is becoming increasingly relevant in this space, it fails to capture the full biophysical complexity of neurons, their nonlinear voltage dynamics, and variability. To address these shortcomings, we introduce NOBLE,...

[182] [Modeling turbulent and self-gravitating fluids with Fourier neural operators](#)

Keith Poletti, ..., and Rachel A. Ward. APL Machine Learning, 2025. 2 citations.

Not measured Topic Match

Abstract: Neural Operators (NOs) are a leading method for surrogate modeling of partial differential equations. Unlike traditional neural networks, which approximate individual functions, NOs learn the mappings between function spaces. While NOs have been predominantly tested on simplified 1D and 2D problems, such as those explored in prior works, these studies fail to address the complexities of more realistic, high-dimensional, and high-dynamic range systems. Moreover, many real-world applications involve incomplete or noisy data, which has not been adequately explored in current NO literature. In this work, we present a novel application of NOs to astrophysical data, which involves high-dynamic range projections...

[183] [A novel hybrid neural network of fluid-structure interaction prediction for two cylinders in tandem arrangement](#)

Yanfeng Lyu, ..., and Yongmao Pei. Engineering Applications of Computational Fluid Mechanics, 2025. 6 citations.

Not measured Topic Match

Abstract: Fluid-structure interaction (FSI) in multibody systems, a non-negligible phenomenon in engineering applications, has been extensively studied via traditional experimental and simulation methods with high cost and time consumption. Deep learning has shown promise in improving computing efficiency while ensuring modelling accuracy in FSI analysis. However, its current capabilities are limited when it comes to constructing multi-object coupling systems with dynamic boundaries. In this paper, we propose a novel FSI hybrid neural network solver integrated by an innovative fluid deep learning model and the structural motion equations for the vortex-induced vibration (VIV) modelling of two tandem cylinders. This well-designed solver, in...

[184] [Enhancing Fourier Neural Operators with Local Spatial Features](#)

Chaoyu Liu, ..., and C. Schönlieb. ArXiv, 2025. 12 citations.

Not measured Topic Match

Abstract: Partial Differential Equation (PDE) problems often exhibit strong local spatial structures, and effectively capturing these structures is critical for approximating their solutions. Recently, the Fourier Neural Operator (FNO) has emerged as an efficient approach for solving these PDE problems. By using parametrization in the frequency domain, FNOs can efficiently capture global patterns. However, this approach inherently overlooks the critical role of local spatial features, as frequency-domain parameterized convolutions primarily emphasize global interactions without encoding comprehensive localized spatial dependencies. Although several studies have attempted to address this limitation, their extracted Local Spatial Features (LSFs) remain insufficient, and computational efficiency is often...

[185] [Mitigating Spectral Bias in Neural Operators via High-Frequency Scaling for Physical Systems](#)

Siavash Khodakarami, ..., and G. Karniadakis. Neural networks : the official journal of the International Neural Network Society, 2025. 36 citations.

Not measured Topic Match

Abstract: Neural operators have emerged as powerful surrogates for modeling complex physical problems. However, they suffer from spectral bias making them oblivious to high-frequency modes, which are present in multiscale physical systems. Therefore, they tend to produce over-smoothed solutions, which is particularly problematic in modeling turbulence and for systems with intricate patterns and sharp gradients such as multi-phase flow systems. In this work, we introduce a new approach named high-frequency scaling (HFS) to mitigate spectral bias in convolutional-based neural operators. By integrating HFS with proper variants of UNet, we demonstrate a higher prediction accuracy by mitigating spectral bias in single and...

[186] [Neural operator surrogate models of plasma edge simulations: feasibility and data efficiency](#)

N. Carey, ..., and P. Setinek. Nuclear Fusion, 2025. 6 citations.

Not measured Topic Match

Abstract: Simulation-based plasma scenario development plays a crucial role in designing next-generation tokamaks and fusion power plants. However, the inclusion of high-fidelity simulations of scrape-off layer turbulence and transient MHD events such as edge localized modes in highly iterative applications remains computationally prohibitive, limiting their use in design and control workflows. Understanding these phenomena is vital, as they govern heat flux on plasma-facing components, influencing reactor performance and material lifetime. This study explored Fourier neural operators (FNOs) as surrogate models to accelerate plasma simulations from the JOEKE MHD and STORM turbulence codes. FNOs were trained on single-step rollouts and evaluated in...

[187] [A general reduced-order neural operator for spatio-temporal predictive learning on complex spatial domains](#)

Qinglu Meng, ..., and Xiaozhong Hao. ArXiv, 2024. 5 citations.

Not measured Topic Match

Abstract: Predictive learning for spatio-temporal processes (PL-STP) on complex spatial domains plays a critical role in various scientific and engineering fields, with its essence being the construction of operators between infinite-dimensional function spaces. This paper focuses on the unequal-domain mappings in PL-STP and categorising them into increase-domain and decrease-domain mapping. Recent advances in deep learning have revealed the great potential of neural operators (NOs) to learn operators directly from observational data. However, existing NOs require input space and output space to be the same domain, which pose challenges in ensuring predictive accuracy and stability for unequal-domain mappings. To this end, this...